U.U.D.M. Project Report 2025:17

Examensarbete C, 15 hp
Kandidatprogrammet i matematik

Juni 2025

Can one hear the area of a drum?

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ABSTRACT

This thesis focuses on proving Weyl's law for arbitrary, two-dimensional domains. The title is inspired by the expression popularized by Mark Kac in his article "Can one hear the shape of a drum?" The considered domains are characterized such that they resemble the physical structure of membranes with clamped endpoints, where the boundary can attain any shape as long as it is smooth. By considering the wave equation for the perpendicular displacements of any point of a domain, it is derived that the constituted homogeneous Dirichlet eigenvalue problem yields a spectra of eigenvalues that is correlated to the domain's vibrational frequencies. Consequently, some of the classical results of the Laplacian and the related eigenfunctions and eigenvalues are highlighted and then utilized in the context of variational calculus to find upper and lower bounds to spectra of different shapes. The homogeneous Dirichlet and Neumann eigenvalue problems are also solved for some particular shapes known to have explicit solutions. These results are then used to approximate arbitrary domains and to lastly prove Weyl's law, which describes an asymptotic relationship between the spectra and the area of a domain. While the result can be extended to other dimensions, this thesis only answers the question in the affirmative for the two-dimensional case.

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1. INTRODUCTION

In April 1966, the mathematician Mark Kac submitted an article in the 73rd volume of *The American Mathematical Monthly* titled "Can one hear the shape of a drum?" [6]. While Kac did not provide a definitive answer to the question (and it wouldn't be answered in the negative until 1992 [5]) he did cover some well-known, related results and further popularized the problem through the article's publication. One of these results was *Weyl's law* - an asymptotic relation first proven in 1911 by Hermann Weyl [13] that instead induced the similar question "can one hear the *area* of a drum?". Given the title of this thesis, answering this question will be our focal point.

1.1. Physical background

Before discussing drums, let us start by introducing some basic concepts related to sound. *Sound waves* that propagate through a medium (air, water, wood etc.) are perceived by the human ear as sound. These are mechanical waves that can be decomposed into sums of sinusoidal waves of different frequencies, which in turn are perceived as *pure tones*. Lower frequencies correspond to lower tones (or lower *pitches*, in musical terminology) while higher frequencies correspond to higher pitches. We can thus informally say that these different frequencies determine the sound we hear.

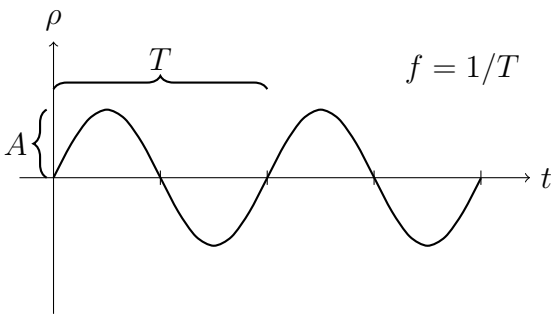


Figure 1.1: A sinusoidal wave in the $t\rho$-plane, where ρ = air pressure, t = time, T = period, A = amplitude and f = frequency. Note that in the case of a drum being struck once, the amplitude decays over time.

1.2. The wave equation

For our purposes, let $\Omega \subsetneq \mathbb{R}^2$ be a non-empty, open and connected domain with a smooth boundary $\partial\Omega$. If the boundary is fixed, this domain can be interpreted as a membrane with clamped endpoints, such as a drum.

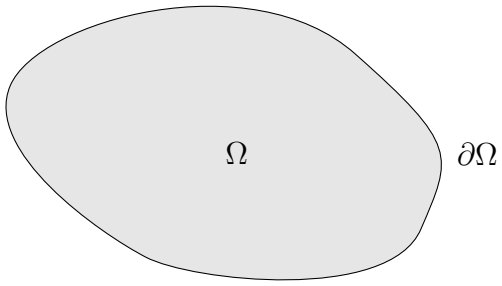


Figure 1.2

Now, let $(x, y) \in \overline{\Omega}$ be any point on the drum including the boundary. When the membrane vibrates, the perpendicular displacement $p(t, x, y) \in C^2([0, \infty) \times \overline{\Omega})$ of any such point with respect to the membrane's surface obeys the two-dimensional *wave equation* with homogeneous *Dirichlet* boundary conditions:

$$\begin{cases} \dfrac{\partial^2 p}{\partial t^2} = c^2 \Delta p & (x, y) \in \Omega, t > 0 \\ p = 0 & (x, y) \in \partial\Omega, t > 0 \\ p = p_1 & (x, y) \in \Omega, t = 0 \\ \dfrac{\partial p}{\partial t} = p_2 & (x, y) \in \Omega, t = 0, \end{cases} \tag{1.1}$$

where $c \in \mathbb{R}$ is the propagation speed of the wave [12], $t \in [0, \infty)$ represents time and $p_1(x, y), p_2(x, y)$ are given initial conditions. For the sake of simplicity, we'll assume $c = 1$, since the inclusion of an arbitrary c would not affect the results of this thesis. Using the *method of separation of variables*, we make the ansatz $p(t, x, y) = \tau(t)u(x, y)$. Inserting this into the PDE of (1.1) yields:

$$\frac{d^2\tau(t)}{dt^2} u(x, y) = \tau(t) \Delta u(x, y) \quad \Longrightarrow \quad \frac{d^2\tau(t)}{dt^2} \frac{1}{\tau(t)} = \frac{\Delta u(x, y)}{u(x, y)}, \tag{1.2}$$

where $(x, y) \in \Omega$ and $t > 0$. Since the two sides of (1.2) depend on different variables, these should both equal some value $-\lambda \in \mathbb{R}$. It is necessarily negative as the force

acting on the drum decays the vibrations over time. This yields the following systems of differential equations:

$$\begin{cases} \dfrac{d^2\tau}{dt^2} + \lambda\tau = 0 & t > 0 \\ \tau = \tau_1 & t = 0 \\ \dfrac{d\tau}{dt} = \tau_2 & t = 0, \end{cases} \tag{1.3}$$

$$\begin{cases} \Delta u + \lambda u = 0 & (x, y) \in \Omega \\ u = 0 & (x, y) \in \partial\Omega. \end{cases} \tag{1.4}$$

We have effectively transformed our initial problem into two so called *Sturm-Liouville* problems. In particular, the spatial system (1.4) will be referred to as the *Dirichlet eigenvalue problem*. Without specifying the boundary condition $u = 0$, the problem can more generally be expressed as

$$\begin{cases} \Delta u + \lambda u = 0 & (x, y) \in \Omega \\ B[u] = 0 & (x, y) \in \partial\Omega, \end{cases} \tag{L}$$

where $B[u]$ represents either homogeneous Dirichlet or *Neumann* boundary conditions. We will refer to (L) as the *Laplacian eigenvalue problem*. Non-trivial solutions to (L) are called *eigenfunctions* $u_i(x, y)$ with corresponding *eigenvalues* λ_i ($i \in \mathbb{N}_{\geq 1}$). In particular, if the domain Ω satisfies some specific properties, there exists a unique, discrete infinite sequence of eigenfunctions $\{u_i\}_{i=1}^{\infty}$ and eigenvalues $\{\lambda_i\}_{i=1}^{\infty}$ solving $\Delta u + \lambda u = 0$ with some boundary condition $B[u]$ [3]. This sequence of eigenvalues is called the domain's *spectra* and will be denoted $\sigma(\Omega)$.

Our temporal equation system (1.3) is a 2nd-order ODE and has the general solution

$$\tau(t) = c_1 \cos(\sqrt{\lambda}t) + c_2 \sin(\sqrt{\lambda}t), \tag{1.5}$$

where the constants c_1, c_2 are determined by the initial conditions [3]. Note that

$$\frac{\sqrt{\lambda}}{2\pi} = f,$$

where f is the frequency of one of the pure tones the membrane Ω is capable of producing [6]. Pairing this observation with the solutions to (L), we can conclude that the eigenvalues of the spectra $\sigma(\Omega)$ are in unique correspondence with the pure tones f_i.

1.3. The main goal

Since the eigenvalues are related to the frequencies of the drum, we would like to deduce a particular relation between the eigenvalues and the area $A(\Omega)$ for a certain category of domains Ω. This relation is the aforementioned Weyl's law and our aim will be to prove it. I won't state Weyl's law in this introduction, so for anyone interested in seeing this asymptotic relation right away, see Section 5.2.

To start, some mathematical groundwork will be necessary to draw some fundamental conclusions about the properties of the considered domains Ω, the *Laplacian* Δ and the related eigenfunctions and eigenvalues that solve (L).

1.4. Notational remarks

Some notational liberties will be taken to facilitate the reading experience. These are valid throughout unless otherwise specified:

1. The natural numbers $\mathbb{N}$ are assumed to exclude 0, ergo
$$\mathbb{N} := \{1, 2, 3, ...\}.$$
2. The letters n, k, i, j will be used extensively as indexing numbers. In particular, we will assume that $n, k, r \in \mathbb{N}\backslash\{1\}$ while the values of i, j will be evident in the context in which they're used.
3. Any mention of a function f belonging to some space of functions in $\mathbb{R}^2$ will be assumed to depend on the variables $(x, y) \in \mathbb{R}^2$, i.e.
$$f = f(x, y).$$
4. The abbreviations "LHS" and "RHS" stand for "left hand side" and "right hand side" respectively.

2.1. Properties of the domain Ω

The membranes of realistic drums are almost exclusively circular. Nonetheless, in the context of mathematical theory, we'll allow ourselves to consider any arbitrary shape in the plane, such as triangles, rectangles and even completely irregular shapes that lack any form of symmetry. Therefore, the question "can one hear the area of a drum" is really asking "can one hear the area of any 2-dimensional membrane". Note that even Kac remarked in his article that the question was more analogous to a tambourine than to a drum [6].

Based on the specifics given under Section 1.2, we define the particular types of domains we're referring to when we write Ω.

Definition 2.1. The domain $\Omega \subsetneq \mathbb{R}^2$ is non-empty, open and connected with a smooth boundary $\partial\Omega$.

Most of these characteristics follow from the nature of drums and are thus fairly self-explanatory. Formal definitions of the topological notions can nonetheless be found in [11]. It should be noted, however, that the criterion of a smooth boundary is of course not applicable to domains with vertices. To avoid any confusion, we'll therefore refer to the domains with vertices in Chapter 4 with indexes that are unique to the respective shapes. The absence of vertices for when we consider *arbitrary* domains Ω will in turn be briefly commented on toward the end of this thesis.

2.2. Boundary conditions

Consider the Laplacian eigenvalue problem given by (L). If $B[u] = u$, we have the homogeneous Dirichlet boundary condition as in (1.4), and if $B[u] = \frac{\partial u}{\partial \vec{n}}$, where $\vec{n}$ is the unit outer normal on $\partial\Omega$, we have the homogeneous Neumann boundary condition. Even though the physical notion of drums is more related to the Dirichlet case, the Neumann condition will be necessary for comparative reasons later on. Consequently, many of the theorems throughout this thesis will be concerned with both the Dirichlet

and the Neumann cases, and some remarks on their differences will be highlighted along the way. The notation $B[u]$ will thus *only* refer to either homogeneous Dirichlet or homogeneous Neumann conditions.

2.3. Inner product spaces

For anyone who has read a course of linear algebra, the notions of eigenfunctions and eigenvalues are based on the equation given by

$$A[v] = \lambda v, \tag{2.1}$$

where A is a linear operator acting on a vector space V and any solution $v \in V$ is called an *eigenvector* with an associated eigenvalue λ. The set of eigenvectors $\{v_i\}$ satisfying $A[v] = \lambda v$ constitutes a linear subspace of V, where the dimension is determined by the *algebraic multiplicity* of λ (i.e. the number of distinct eigenfunctions that solve (2.1) for the same eigenvalue). Familiarly, the notions of *inner products* and their properties are applicable to vectors of linear spaces and will be defined based on [14] and [8] as follows.

Definition 2.2. An *inner product space* is a pair $(V, \langle \cdot, \cdot \rangle)$, where V is a linear space and $\langle u, v \rangle \in \mathbb{C}$ is an inner product of any two arbitrary vectors $u, v \in V$ which satisfies the following properties:

1. $\langle u, v \rangle = \overline{\langle v, u \rangle}$ $\forall u, v \in V$.
2. $\langle u + w, v \rangle = \langle u, v \rangle + \langle w, v \rangle$ $\forall u, v, w \in V$.
3. $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$ $\forall u, v \in V$ and $\forall \alpha \in \mathbb{C}$.
4. $\langle v, v \rangle \geq 0$ $\forall v \in V$ such that $\langle v, v \rangle = 0 \iff v = 0$.

Definition 2.3. Each inner product has an induced *norm* defined as $||v|| := \langle v, v \rangle^{1/2}$ with the following properties:

1. $||\alpha v|| = |\alpha| \cdot ||v||$ $\forall v \in V$ and $\forall \alpha \in \mathbb{C}$.
2. $||u + v|| \leq ||u|| + ||v||$ $\forall u, v \in V$.
3. $||v|| \geq 0$ $\forall v \in V$ such that $||v|| = 0 \iff v = 0$.
4. $| \langle u, v \rangle | \leq ||u|| \cdot ||v||$ $\forall u, v \in V$.

Definition 2.4. Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and let $\{u_i\} \subset V$ be an ordered sequence of vectors. Then the following properties hold:

1. If $\langle u_i, u_j \rangle = 0$ for $i \neq j$, then the vectors u_i, u_j are called *orthogonal*.
2. If $u_i \neq 0 \ \forall i$ and $\langle u_i, u_j \rangle = 0 \ \forall i \neq j$, then the sequence $\{u_i\}$ is called orthogonal.
3. If
$$\langle u_i, u_j \rangle = \begin{cases} 0 & i \neq j \\ 1 & i = j, \end{cases}$$
then the sequence $\{u_i\}$ is called *orthonormal*.

Remark 2.1. Note that two vectors $u, v \in V$ are orthonormal if and only if they are orthogonal and satisfy $||u|| = ||v|| = 1$ by this definition.

2.4. Properties of the Laplacian, eigenfunctions and eigenvalues

The notions of inner product spaces are of relevance to our problem $\Delta u + \lambda u = 0$ since the Laplacian is also a linear operator. In particular, the Laplacian is a *2nd-order differential operator* with the following definition.

Definition 2.5. The Laplacian Δ is an operator satisfying
$$\Delta : C^n(\mathbb{R}^2) \longrightarrow C^{(n-2)}(\mathbb{R}^2),$$
$$u \longmapsto \Delta u,$$
where Δu is given by
$$\Delta u = \nabla \cdot \nabla u = \nabla \cdot \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}.$$

We pair this with the following definition for the inner product. It can be checked that it satisfies the criteria given in Definition 2.2.

Definition 2.6. The inner product of any two functions $u, v \in L^2(\Omega)$ is defined as:
$$\langle u, v \rangle := \iint_\Omega u(x, y) \overline{v(x, y)} dx dy. \tag{2.2}$$

Remark 2.2. We will allow for inner products between any two vector fields $\mathbf{u}, \mathbf{v}$ to also be written as $\langle \mathbf{u}, \mathbf{v} \rangle$ for convenience sake.

2.4.1. Classical and weak solutions

Solutions to the Laplacian eigenvalue problem that are well-defined in all of $\overline{\Omega}$ satisfy the following definition.

Definition 2.7. Consider the Laplacian eigenvalue problem (L) and assume u and $\tilde{u}$ are solutions to the Dirichlet and Neumann problems respectively. Then we call the solutions *classical* if they are unique and satisfy

$$u \in C^2(\Omega) \cap C(\overline{\Omega}),$$
$$\tilde{u} \in C^2(\Omega) \cap C^1(\overline{\Omega}).$$

Remark 2.3. $C(\overline{\Omega}) \subset L^2(\Omega)$ since Ω is bounded, making the inner product given in Definition 2.6 valid for these solutions.

We say that a given PDE is solved in the classical sense if one manages to find the classical solution. However, for the Laplacian eigenvalue problem, this is only possible for a few number of domains with certain symmetrical properties (we'll solve some of these in Chapter 4). Actually, most other PDEs cannot be solved in the classical sense [3] and demand for wider classes of functions to be considered. Solutions of this type are called *weak* solutions and are defined as follows.

Definition 2.8. Solutions to the Laplacian eigenvalue problem that are not classical in accordance to Definition 2.7 are called weak solutions and satisfy

$$u \in H_0^1(\Omega),$$

$$\tilde{u} \in H^1(\Omega)$$

for the respective boundary conditions.

The spaces H_0^1, H^1 are known as *Sobolev spaces* and are equipped with a norm different from the one induced by Definition 2.6. The properties of these spaces and the motivation for their use will not be discussed in detail here (I suggest [3] for anyone interested), but their implementation will allow us to justify some classifications of particular functions at the start of Chapter 3. Regardless, we can assume until then that a "solution" to (L) satisfies the same smoothness properties that are implied by Definition 2.7.

2.4.2. Green's identities

We will need *Green's first and second identity* in many of our calculations. Both of these are based on the *Gauss-Green Theorem* (A.1), which we'll provide the two-dimensional version of here:

Theorem 2.1. *Consider any vector field* $\boldsymbol{u} \in C^1(\Omega;\mathbb{R}^2) \cap C(\overline{\Omega};\mathbb{R}^2)$. *Then*

$$\iint_\Omega \nabla \cdot \boldsymbol{u}\, dxdy = \int_{\partial\Omega} \boldsymbol{u} \cdot \vec{n}\, dS, \tag{G}$$

where $\vec{n}$ *is the unit outer normal on* $\partial\Omega$ *and* dS *is the surface element on* $\partial\Omega$.

Proposition 2.1. *Consider any two functions* $u, v \in C^2(\Omega) \cap C(\overline{\Omega})$. *Then Green's first identity is given by*

$$\int_{\partial\Omega} v\frac{\partial u}{\partial \vec{n}}\, dS = \iint_\Omega \nabla v \cdot \nabla u\, dxdy + \iint_\Omega v\Delta u\, dxdy, \tag{G1}$$

where $\frac{\partial u}{\partial \vec{n}} = \vec{n} \cdot \nabla u$ *is the directional derivative in the outward normal direction.*

Proof. Consider the expression $\nabla\cdot(v\nabla u)$ [note that $(v\nabla u)$ is a two-dimensional vector field]. Using the chain rule, we get

$$\nabla \cdot (v\nabla u) = \nabla v \cdot \nabla u + v\Delta u.$$

By integrating both sides over Ω and utilizing (G), we get

$$\begin{aligned}
&\iint_\Omega \nabla \cdot (v\nabla u)\, dxdy = \iint_\Omega \nabla v \cdot \nabla u + v\Delta u\, dxdy \\
&= \int_{\partial\Omega} (v\nabla u) \cdot \vec{n}\, dS \\
&= \int_{\partial\Omega} v\frac{\partial u}{\partial \vec{n}}\, dS = \iint_\Omega \nabla v \cdot \nabla u\, dxdy + \iint_\Omega v\Delta u\, dxdy.
\end{aligned}$$

This is precisely Green's first identity. □

Conveniently, Green's second identity is directly derived from the first identity.

Proposition 2.2. *Consider any two functions* $u, v \in C^2(\Omega) \cap C(\overline{\Omega})$. *Then Green's second identity is given by*

$$\int_{\partial\Omega} \left(v\frac{\partial u}{\partial \vec{n}} - u\frac{\partial v}{\partial \vec{n}} \right)\, dS = \iint_\Omega (v\Delta u - u\Delta v)\, dxdy. \tag{G2}$$

Proof. We begin by noting that $\nabla v \cdot \nabla u = \nabla u \cdot \nabla v$ for any u, v. Thus, by utilizing (G1), we have

$$\begin{aligned} \int_{\partial\Omega} v\frac{\partial u}{\partial \vec{n}} - u\frac{\partial v}{\partial \vec{n}}\, dS &= \iint_{\Omega} \nabla v \cdot \nabla u - \nabla u \cdot \nabla v\, dxdy + \iint_{\Omega} v\Delta u - u\Delta v\, dxdy \\ \implies \int_{\partial\Omega} v\frac{\partial u}{\partial \vec{n}} - u\frac{\partial v}{\partial \vec{n}}\, dS &= \iint_{\Omega} v\Delta u - u\Delta v\, dxdy. \end{aligned}$$

This is precisely Green's second identity. □

We'll now provide a theorem that covers some of the most fundamental properties of the eigenvalues related to (L). These will be of central importance and are either justified or commented on continuously throughout the rest of this thesis.

Theorem 2.2. *Consider the Laplacian eigenvalue problem* (L) *for the domain* Ω *with either Dirichlet or Neumann boundary conditions. Let* $\{u_j\}$ *be the set of all eigenfunctions (counted with multiplicity) and let* $\{\lambda_i\}$ *be the ordered set of associated eigenvalues. Then the following properties hold:*

1. *All eigenvalues are real.*
2. *The set* $\{\lambda_i\}$ *is unbounded, monotonically non-decreasing and is sequenced as*

$$0 < \lambda_1 < \lambda_2 \leq \lambda_3 \leq \cdots \to \infty,$$

for the Dirichlet boundary condition and

$$0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \cdots \to \infty.$$

for the Neumann boundary condition.

We'll continue by proving some of these results right away.

2.4.3. Symmetry

Theorem 2.3. *Consider any two eigenfunctions* u, v *solving the Laplacian eigenvalue problem with either boundary condition. Then the Laplacian* Δ *satisfies*

$$\langle \Delta u, v \rangle = \langle u, \Delta v \rangle$$

and is called symmetric/self-adjoint.

Proof. If we consider the LHS of (G2), it is clear that the integrand disappears at the boundary for either Dirichlet or Neumann boundary conditions. We thus have

$$\int_{\partial\Omega} \left(v\frac{\partial u}{\partial \vec{n}} - u\frac{\partial v}{\partial \vec{n}} \right) \, dS = 0.$$

The resulting identity of (G2) becomes

$$0 = \iint_{\Omega} (v\Delta u - u\Delta v) \, dxdy \quad \Longrightarrow \quad \iint_{\Omega} v\Delta u \, dxdy = \iint_{\Omega} u\Delta v \, dxdy.$$

By Definition 2.6, this is the same as $\langle \Delta u, v \rangle = \langle u, \Delta v \rangle$. □

2.4.4. Orthogonality

Proposition 2.3. *Any two eigenfunctions u, v solving the Laplacian eigenvalue problem with either boundary condition are orthogonal to each other if they are associated with different eigenvalues. Moreover, the respective gradients of two such eigenfunctions are also orthogonal.*

Proof. Let $\lambda_u \neq \lambda_v$ denote the eigenvalues of the respective eigenfunctions u, v, i.e.,

$$\begin{aligned} -\Delta u &= \lambda_u u, \\ -\Delta v &= \lambda_v v. \end{aligned}$$

Using the symmetry property of Theorem 2.3 and substituting with the identities above, we get

$$0 = \iint_{\Omega} (v\Delta u - u\Delta v) \, dxdy = \iint_{\Omega} (-v\lambda_u u + u\lambda_v v) \, dxdy = (\lambda_v - \lambda_u) \iint_{\Omega} uv \, dxdy.$$

Since $\lambda_v - \lambda_u \neq 0$, it follows that $\langle u, v \rangle = 0$. Thus, the eigenfunctions are orthogonal.

If we now consider the inner product given by $\langle \nabla u, \nabla v \rangle$ for the same pair u, v, we get by (G1) that

$$\begin{aligned} \langle \nabla u, \nabla v \rangle &= \iint_{\Omega} \nabla u \cdot \nabla v \, dxdy \\ &= \int_{\partial\Omega} u\frac{\partial v}{\partial \vec{n}} \, dS - \iint_{\Omega} u\Delta v \, dxdy. \end{aligned} \tag{2.3}$$

Note that the term $\int_{\partial\Omega} u\frac{\partial v}{\partial \vec{n}}\, dS = 0$ for either boundary condition $B[u]$. Substituting this into (2.3) yields

$$\begin{aligned}\langle \nabla u, \nabla v\rangle &= -\iint_\Omega u\Delta v\ dxdy \\ &= \iint_\Omega u\lambda_v v\ dxdy \\ &= \lambda_v \iint_\Omega uv\ dxdy \\ &= \lambda_v \langle u, v\rangle = 0.\end{aligned}$$

This confirms that if $\langle u, v\rangle = 0$, then $\langle \nabla u, \nabla v\rangle = 0$. □

2.4.5. Real eigenvalues

Note from Definition 2.5 that the Laplacian Δ is a real operator. This observation leads to the following proposition.

Proposition 2.4. *The eigenvalues of the Laplacian eigenvalue problem are real.*

Proof. We'll prove our proposition by contradiction. Assume $\exists\lambda \in \mathbb{C}\backslash\mathbb{R}$ with an associated eigenfunction u. With this assumption, we have $\lambda \neq \bar{\lambda}$. By taking the complex conjugate [11] of (L), we get

$$\begin{cases}\overline{\Delta u + \lambda u} = \Delta\bar{u} + \bar{\lambda}\bar{u} = 0 & (x, y) \in \Omega \\ \overline{B[u]} = B[\bar{u}] = 0 & (x, y) \in \partial\Omega.\end{cases}$$

This confirms that $\bar{u}$ is also an eigenfunction associated with the eigenvalue $\bar{\lambda}$ solving (L) with either boundary condition. By Proposition 2.3, the two eigenfunctions are orthogonal to each other and thus satisfy

$$0 = \iint_\Omega u\bar{u}\ dxdy = \langle u, u\rangle .$$

However, this leads to a contradiction, since $u \neq 0 \Longrightarrow \langle u, u\rangle > 0$ by Definition 2.2. □

2.4.6. The Rayleigh Quotient

Consider the Laplacian eigenvalue problem given by (L) with either boundary condition. If we multiply both sides with u and integrate over Ω, we get

$$(-\Delta u) \cdot u = \lambda |u|^2 \quad \Longrightarrow \quad \iint_\Omega (-\Delta u) \cdot u \; dxdy = \lambda \iint_\Omega |u|^2 \; dxdy. \tag{2.4}$$

Utilizing (G1) on the LHS of (2.4) and applying either boundary condition gives

$$\begin{aligned} &\iint_\Omega (-\Delta u) \cdot u \; dxdy \\ &= \iint_\Omega |\nabla u|^2 \; dxdy = \lambda \iint_\Omega |u|^2 \; dxdy \\ &\iff \lambda = \frac{\iint_\Omega |\nabla u|^2 \; dxdy}{\iint_\Omega |u|^2 \; dxdy}. \end{aligned}$$

By Definition 2.3 and Definition 2.6, this can be rewritten as

$$\lambda = \frac{||\nabla u||^2}{||u||^2}. \tag{2.5}$$

The RHS of (2.5) is known as the *Rayleigh quotient* and is often utilized in variational problems in this area of math. We will use the Rayleigh quotient extensively in Chapter 3, so a definition here is in order.

Definition 2.9. For the Laplacian eigenvalue problem of Ω with either boundary condition, let u denote any eigenfunction. Then, the Rayleigh quotient $R(u)$ is defined as

$$R(u) := \frac{||\nabla u||^2}{||u||^2}, \tag{R}$$

where $u \in L^2(\Omega) \backslash \{0\}$ and $\nabla u \in L^2(\Omega; \mathbb{R}^2)$. If $u|_{\partial\Omega} = 0$, then $\nabla u \neq (0, 0)$ additionally.

The extra constraint on ∇u for the Dirichlet boundary condition in this definition is justified by the following proposition.

2.4.7. Positive eigenvalues

Proposition 2.5. *The eigenvalues associated with the Laplacian eigenvalue problem are*

1. *strictly positive for Dirichlet boundary conditions.*

2. non-negative for Neumann boundary conditions.

Proof of 1. Considering (R), it is obvious that $\lambda \geq 0$. Furthermore, since $u = 0$ on $\partial\Omega$, this implies that there cannot exist constant eigenfunctions. Thus, $\nabla u \neq (0, 0)$, and so $\lambda > 0$. □

Proof of 2. Considering (R) again, the boundary condition $\frac{\partial u}{\partial \vec{n}} = 0$ on $\partial\Omega$ doesn't negate the possibility of there existing a constant eigenfunction. Thus, $\lambda \geq 0$. □

2.4.8. Further properties

While we've proven many of the central components of Theorem 2.2, there remain some further properties that are fundamental in the context of PDE-theory, but slightly beyond the scope of this thesis. We claimed in Section 1.2 that there exists a unique, discrete infinite sequence of eigenfunctions $\{u_i\}_{i=1}^{\infty}$ of the Laplacian eigenvalue problem. It can be shown that this sequence, in fact, admits a real, *orthonormal basis* of $L^2(\Omega)$, i.e.

$$\overline{\text{span}}\{u_i\}_{i=1}^{\infty} = L^2(\Omega).$$

This justifies the implementation of the separation of variables method used to solve the wave equation (1.1), as any element $l \in L^2(\Omega)$ can be expressed as

$$l = \sum_{i=1}^{\infty} \langle u_i, l \rangle \, u_i.$$

Since we won't justify these claims, I refer the interested reader to [3].

3. APPLICATIONS OF VARIATIONAL CALCULUS

Now we embark on a small section of the extensive subject of the calculus of variations. Going forward, the eigenfunctions/-values of the Dirichlet problem will be denoted as u_i and λ_i, while the ones related to the Neumann problem will be denoted as $\tilde{u}_i$ and $\tilde{\lambda}_i$ *unless otherwise specified*.

3.0.1. Minimizing the Rayleigh quotient

We will in this chapter try and find minima of the Rayleigh quotient R given in Definition 2.9 over some class of functions. Given that such a function is chosen suitably, this minimum should be some eigenvalue and the minimizer should be the associated eigenfunction of the Laplacian eigenvalue problem. The aforementioned weak solutions of Definition 2.8 are suitable for this purpose since they are well-defined for R and the Sobolev spaces to which they belong are complete [3]. Furthermore, by studying the smoothness properties of these weak solutions using *regularity theory* [3], one can conclude that they belong to the class $C^\infty(\Omega) \cap C^i(\overline{\Omega})$, where i reflects the regularity of the boundary $\partial\Omega$. Since we've assumed that the boundary is smooth by Definition 2.1, this space is precisely $C^\infty(\overline{\Omega})$. This important discovery gives us a lot of freedom for when we choose what spaces of functions we want to minimize R over and still expect a valid result.

3.0.2. Trial functions

When we extracted the Rayleigh quotient in Section 2.4.6, we assumed that the given λ and u satisfied the Laplacian eigenvalue problem (L), but we cannot assume the contrary holds - i.e. if we are given $R(u)$, then how do we know that u is an eigenfunction and that the minimum of $R(u)$ is an associated eigenvalue? Any such u will thus be referred to as a *trial function* until proven to be an eigenfunction. We will have different trial functions v, w for the Dirichlet and Neumann cases and will assume that they satisfy

$$v \in C_c^\infty(\Omega)\backslash\{0\},$$
$$w \in C^\infty(\overline{\Omega})\backslash\{0\}.$$

Both of these classes are subsets of $C^\infty(\overline{\Omega})$ and are chosen to best suit our purposes (in ways that'll become apparent later). Note that the *compact support* (A.3) of the class $C_c^\infty(\Omega)$ implies the Dirichlet boundary condition $v|_{\partial\Omega} = 0$. In contrast, the Neumann trial functions lack any homogeneous boundary conditions. We'll explain this deviation later on.

3.1. The Minimum principles

The Laplacian eigenvalue problem given by (L) will be considered throughout the rest of this chapter. Additionally, the notation for the integration variables will be omitted for the sake of readability. The different proofs and conclusions that follow are based on content in [12].

Recall from Theorem 2.2 that the set of eigenvalues are ordered as such:

$$0 < \lambda_1 < \lambda_2 \leq \lambda_3 \leq ... \to \infty,$$
$$0 = \tilde{\lambda}_1 < \tilde{\lambda}_2 \leq \tilde{\lambda}_3 \leq ... \to \infty.$$

The first eigenvalues λ_1, $\tilde{\lambda}_1$ are called the *principal eigenvalues* and are associated to the principal eigenfunctions $u_1, \tilde{u}_1$ respectively.

3.1.1. The Dirichlet eigenvalues λ_i

Theorem 3.1. *The principal eigenvalue λ_1 for the Dirichlet eigenvalue problem of the domain Ω is given by*

$$\lambda_1 = \min_{v \in V} \frac{||\nabla v||^2}{||v||^2}, \tag{3.1}$$

where $V := \{v \in C_c^\infty(\Omega) \backslash \{0\}\}$. The associated trial function v satisfying (3.1) *is the principal eigenfunction u_1.*

We need the following lemma in order to prove this theorem. This is a two-dimensional version of a lemma found in [4].

Lemma 3.1. *If $f(x,y) \in C(\overline{\Omega})$ and if*

$$\iint_\Omega f(x,y)g(x,y) = 0$$

for every function $g(x,y) \in C(\Omega)$ such that $g(x_0,y_0) = 0$ $\forall (x_0,y_0) \in \partial\Omega$, then $f(x,y) = 0$ $\forall (x,y) \in \overline{\Omega}$.

Proof of lemma. Assume $f(x,y)$ is positive at some point in $\overline{\Omega}$. Then $f(x,y)$ is also positive in some region $\Omega' \subset \Omega$. If we choose any $g(x,y) \in C_c(\Omega')$ such that $g \geq 0\ \forall (x,y) \in \overline{\Omega}$ (which satisfies the conditions of the lemma), we get

$$\iint_\Omega f(x,y)g(x,y) = \iint_{\Omega'} f(x,y)g(x,y) > 0.$$

This contradicts our first assumption. The result becomes the same if we'd chosen $f(x,y)$ to be negative at some point in $\overline{\Omega}$ instead. □

Proof of theorem. Let $m := \min\limits_{v \in V} \frac{||\nabla v||^2}{||v||^2}$ and assume the trial function $v_1 \in V$ solves (3.1), i.e.

$$m = \frac{||\nabla v_1||^2}{||v_1||^2} \leq \frac{||\nabla v||^2}{||v||^2}$$

for any $v \in V$. Let $\phi \in V$ be any other trial function such that $v = v_1 + \epsilon\phi$, where $\epsilon \in \mathbb{R}$. Then the function

$$f(\epsilon) = \frac{||\nabla(v_1 + \epsilon\phi)||^2}{||v_1 + \epsilon\phi||^2} = \frac{\iint_\Omega (|\nabla v_1|^2 + 2\epsilon \nabla v_1 \cdot \nabla\phi + \epsilon^2 |\nabla\phi|^2)}{\iint_\Omega (v_1 + 2\epsilon v_1\phi + \epsilon^2\phi^2)}$$

attains its minimum at $\epsilon = 0$, meaning that $\frac{d}{d\epsilon} f(0) = 0$. Note that this is because v_1 is defined such that it minimizes the Rayleigh quotient. This leads to

$$0 = \frac{d}{d\epsilon} f(0) = \frac{(\iint_\Omega v_1^2)(2\iint_\Omega \nabla v_1 \cdot \nabla\phi) - (\iint_\Omega |\nabla v_1|^2)(2\iint_\Omega v_1\phi)}{(\iint_\Omega v_1^2)^2}$$
$$\implies \iint_\Omega \nabla v_1 \cdot \nabla\phi = m \iint_\Omega v_1\phi.$$

Since $\phi|_{\partial\Omega} = 0$, (G1) yields

$$\iint_\Omega \nabla v_1 \cdot \nabla\phi = -\iint_\Omega \Delta v_1 \phi = m \iint_\Omega v_1\phi \implies \iint_\Omega (\Delta v_1 + m v_1)\phi = 0. \tag{3.2}$$

Because $\phi \in C_c^\infty(\Omega)$, we get by Lemma 3.1:

$$\Delta v_1 + m v_1 = 0 \quad \text{in } \Omega.$$

This proves that m is an eigenvalue with an associated eigenfunction v_1.

In order to show that m is the smallest eigenvalue, assume there exists eigenfunctions u_j and eigenvalues λ_j such that

$$\begin{cases} -\Delta u_j = \lambda_j u_j & (x,y) \in \Omega \\ u_j = 0 & (x,y) \in \partial\Omega, \end{cases} \tag{3.3}$$

where $j \in \mathbb{N}\backslash\{1\}$. By (R) and (G1), we get

$$m \leq \frac{||\nabla u_j||^2}{||u_j||^2} = \frac{\iint_\Omega |\nabla u_j|^2}{\iint_\Omega |u_j|^2} = \frac{\iint_\Omega (-\Delta u_j)\cdot u_j}{\iint_\Omega |u_j|^2} = \frac{\iint_\Omega (\lambda_j u_j)\cdot u_j}{\iint_\Omega |u_j|^2} = \lambda_j.$$

This proves that $m = \lambda_1$ and $v_1 = u_1$. □

Theorem 3.2. *Suppose that the eigenvalues $\lambda_1, ..., \lambda_{n-1}$ are already known with the respective eigenfunctions $u_1, ..., u_{n-1}$ for the Dirichlet eigenvalue problem of the domain Ω. Then the n-th eigenvalue λ_n is given by*

$$\lambda_n = \min_{v \in V^*} \frac{||\nabla v||^2}{||v||^2}, \tag{3.4}$$

where $V^* := \{v \in C_c^\infty(\Omega)\backslash\{0\} \mid \langle v, u_1\rangle = ... = \langle v, u_{n-1}\rangle = 0\}$. The associated function v satisfying (3.4) is the n-th eigenfunction u_n.

Remark 3.1. By the definition of V^*, the extra imposed restriction on the preceding eigenfunctions $u_1, ..., u_{n-1}$ is a direct result of Proposition 2.3: They are all necessarily orthogonal to each other since we don't want the n-th eigenvalue to be associated with any of the eigenspaces imposed by any previous eigenfunctions.

Proof. Let $m^* := \min_{v \in V^*} \frac{||\nabla v||^2}{||v||^2}$ and assume the function $v_n \in V^*$ solves (3.4) for any $v \in V^*$. By a similar fashion to the previous proof, we make the substitution $v = v_n + \epsilon\phi$, where $\phi \in V^*$, and conclude that

$$\iint_\Omega (\Delta v_n + m^* v_n)\phi = 0. \tag{3.5}$$

However, contrary to the last proof, we cannot conclude immediately that $\Delta v_n + m^* v_n = 0$ since ϕ is orthogonal to $u_1, ..., u_{n-1}$. This issue is resolved in the following way:

Let $\psi \in V$ be an arbitrary trial function and make the following substitution:

$$\phi = \psi - \sum_{k=1}^{n-1} c_k u_k, \tag{3.6}$$

where $c_k = \frac{\langle \psi, u_k \rangle}{\langle u_k, u_k \rangle}$. By the Gram-Schmidt process (A.2), $\langle \phi, u_j \rangle = 0 \, \forall j \in \{1, ..., n-1\}$. This new way of expressing ϕ therefore still satisfies all the required constraints such that $\phi \in V^*$. Inserting this substitution into (3.5) yields

$$\begin{aligned} \iint_\Omega (\Delta v_n + m^* v_n)\phi &= \iint_\Omega (\Delta v_n + m^* v_n) \left(\psi - \sum_{k=1}^{n-1} c_k u_k \right) \\ &= \iint_\Omega (\Delta v_n + m^* v_n)\psi - \sum_{k=1}^{n-1} c_k \iint_\Omega (\Delta v_n + m^* v_n) u_k = 0. \end{aligned} \tag{3.7}$$

Now, if we look at the integral in the summation in (3.7), (G2) together with the boundary condition yields

$$\begin{aligned} \iint_\Omega (\Delta v_n + m^* v_n) u_k &= \iint_\Omega v_n (\Delta u_k + m^* u_k) \\ &= (m^* - \lambda_k) \iint_\Omega v_n u_k = 0, \end{aligned} \tag{3.8}$$

where $k \in \{1, ..., n-1\}$. This means that

$$\iint_\Omega (\Delta v_n + m^* v_n)\psi = 0.$$

Analogous to the previous proof, this leads to $\Delta v_n + m^* v_n = 0$ and we conclude that m^* is an eigenvalue with an associated eigenfunction v_n.

We show that m^* is the n-th eigenvalue in a similar way to the previous proof. For eigenfunctions u_j and eigenvalues λ_j satisfying (3.3) such that $j \in \mathbb{N}_{\geq n}$, we have $m^* \leq \lambda_j$. Furthermore, $m^* \neq \lambda_i$ for $i \in \{1, ..., n-1\}$, since the contrary would cause the associated eigenfunctions v_n and u_k to *not* be orthogonal, contradicting the constraints given by V^*. This proves that $m^* = \lambda_n$ and $v_n = u_n$. □

3.1.2. The Neumann eigenvalues $\tilde{\lambda}_i$

The eigenvalues of the Neumann eigenvalue problem can be characterized in a similar way.

Theorem 3.3. *The principal eigenvalue $\tilde{\lambda}_1$ for the Neumann eigenvalue problem of the domain Ω is given by*

$$\tilde{\lambda}_1 = \min_{w \in W} \frac{||\nabla w||^2}{||w||^2}, \tag{3.9}$$

where $W := \{w \in C^\infty(\overline{\Omega}) \backslash \{0\}\}$. The associated trial function w satisfying (3.9) is the principal eigenfunction $\tilde{u}_1$.

Proof. The first few steps are analogous to the proof of Theorem 3.1 until the boundary condition matters. Let $\tilde{m} := \min_{w \in W} \frac{||\nabla w||^2}{||w||^2}$ and assume the function $w_1 \in W$ solves (3.9) for any $w \in W$. Furthermore, let $\phi \in W$ be any other trial function. This yields

$$\iint_\Omega \nabla w_1 \cdot \nabla \phi = \tilde{m} \iint_\Omega w_1 \phi.$$

Using (G1), this can be rewritten as

$$\iint_\Omega (\Delta w_1 + \tilde{m} w_1)\phi = \int_{\partial\Omega} \frac{\partial w_1}{\partial \vec{n}} \phi. \tag{3.10}$$

Similar to the proof of the analogous Dirichlet case, we can choose $\phi \in C_c^\infty(\Omega)$ such that $\Delta w_1 + \tilde{m} w_1 = 0$ in Ω by Lemma 3.1. This causes the LHS of (3.10) to equal 0 for *all* trial functions ϕ. We thus get

$$\int_{\partial\Omega} \frac{\partial w_1}{\partial \vec{n}} \phi = 0.$$

And since $\phi \in W$ lack any boundary conditions, we can choose $\phi|_{\partial\Omega} = \frac{\partial w_1}{\partial \vec{n}}$, meaning that

$$0 = \int_{\partial\Omega} \frac{\partial w_1}{\partial \vec{n}} \phi = \int_{\partial\Omega} \left(\frac{\partial w_1}{\partial \vec{n}} \right)^2 \geq \int_{\partial\Omega} 0 \implies \frac{\partial w_1}{\partial \vec{n}} = 0.$$

We've effectively proven not only that $\tilde{m}$ is an eigenvalue with an associated eigenfunction w_1, but also that the Neumann boundary conditions follows by default without the need to add it as a constraint to W.

Deducing that $\tilde{m} = \tilde{\lambda}_1$ and $w_1 = \tilde{u}_1$ is analogous to the proof for the Dirichlet case and we'll therefore skip it. □

Remark 3.2. Note that this proof also confirms that $\tilde{\lambda}_1 = 0$ and is simple, since the associated eigenfunction $\tilde{u}_1$ is any arbitrary constant. While one can prove that the Dirichlet principal eigenvalue λ_1 is also simple, I've chosen not to justify this as it is slightly out of scope for this thesis.

Theorem 3.4. *Suppose that the eigenvalues $\tilde{\lambda}_1, ..., \tilde{\lambda}_{n-1}$ are already known with the respective eigenfunctions $\tilde{u}_1, ..., \tilde{u}_{n-1}$ for the Neumann eigenvalue problem of the domain Ω. Then the n-th eigenvalue $\tilde{\lambda}_n$ is given by*

$$\tilde{\lambda}_n = \min_{w \in W^*} \frac{||\nabla w||^2}{||w||^2}, \tag{3.11}$$

where $W^* := \{w \in C^\infty(\overline{\Omega}) \backslash \{0\} \mid \langle w, \tilde{u}_1 \rangle = ... = \langle w, \tilde{u}_{n-1} \rangle = 0\}$. The associated function w satisfying (3.11) is the n-th eigenfunction $\tilde{u}_n$.

The proof is omitted as it is analogous to the proof of Theorem 3.2.

3.1.3. The Minimax principle

The *Minimax principle* (referred to as the "Minimum-Maximum property of eigenvalues" in [2]) can be derived from the characterizations of the eigenvalues we've just established. This principle proves that the eigenvalues are monotonically non-decreasing as suggested in Theorem 2.2. However, the so called *Maximin principles* will be more useful for our purposes and we'll therefore focus our attention on proving that instead.

3.2. The Maximin principles

The theorems 3.2 and 3.4 for the n-th eigenvalues for both boundary value problems require knowledge of the first $n-1$ eigenfunctions. The Maximin principles allow us to express any eigenfunction without this necessity, making it very handy for when we work with domains that lack analytical solutions (which are most domains Ω).

Theorem 3.5. *Consider the Dirichlet eigenvalue problem of the domain Ω and let $h_1, ..., h_{n-1}$ be arbitrary, piecewise continuous functions in Ω. Then the n-th eigenvalue λ_n is given by*

$$\lambda_n = \max_{\{h_i\}_{i=1}^{n-1}} \min_{v \in \Upsilon} \frac{||\nabla v||^2}{||v||^2}, \tag{3.12}$$

where $\Upsilon := \{v \in C_c^\infty(\Omega) \backslash \{0\} \mid \langle v, h_1 \rangle = ... = \langle v, h_{n-1} \rangle = 0\}$ defines the set of trial functions.

We should clarify what we mean by piecewise continuous in this context.

Definition 3.1. Functions that are piecewise continuous in Ω are allowed to have a finite number of *jump discontinuities*, i.e. points of discontinuity where both left and right limits exists [8].

We define piecewise continuity in this way since we want to allow for Ω to be divided into a finite number of disjoint subdomains $\Omega_1, ..., \Omega_k$ such that, for each $i \in \{1, ..., k\}$, the restriction of any such function to Ω_i has a continuous extension to $\overline{\Omega}_i$. We're thus allowing for these functions to be possibly discontinuous on the internal boundaries $\partial\Omega_1, ..., \partial\Omega_k$, but continuous otherwise. This enables us to compare the eigenvalues related to the Laplacian eigenvalue problem of any subdomain to the ones related to the entire domain.

Proof. Fix an arbitrary sequence $\{h_i\}_{i=1}^{n-1}$ of piecewise continuous functions in Ω and let $v^*(x, y) = \sum_{j=1}^{n} c_j u_j(x, y)$ be a linear combination of the first n normalized eigenfunctions u_j such that v^* is orthogonal to any h_i. The constants $c_j \in \mathbb{R}$ are thus chosen such that

$$\langle v^*, h_i \rangle = \left\langle \sum_{j=1}^{n} c_j u_j, h_i \right\rangle = \sum_{j=1}^{n} \langle u_j, h_i \rangle c_j = 0.$$

In particular, since the system consists of $n - 1$ equations with n unknowns, there exists some solution $c_1, ..., c_n \in \mathbb{R}$ such that not all c_j are 0. This results in the following relation:

$$\min_{v \in \Upsilon} \frac{||\nabla v||^2}{||v||^2} \leq \frac{||\nabla v^*||^2}{||v^*||^2} = \frac{\sum_{j,k=1}^{n} c_j c_k \langle \nabla u_j, \nabla u_k \rangle}{\sum_{j,k=1}^{n} c_j c_k \langle u_j, u_k \rangle}. \tag{3.13}$$

By Proposition 2.3, (G1), the boundary condition $u_j|_{\partial\Omega} = 0$ and the fact that the eigenfunctions are normalized, we have that

$$\langle u_j, u_k \rangle = \begin{cases} 1 & \text{if } j = k \\ 0 & \text{if } j \neq k \end{cases} \qquad \text{and} \qquad \langle \nabla u_j, \nabla u_k \rangle = \begin{cases} \lambda_j & \text{if } j = k \\ 0 & \text{if } j \neq k, \end{cases}$$

where $j, k = 1, ..., n$. And so, (3.13) can be simplified as such:

$$\frac{\sum_{j,k=1}^{n} c_j c_k \langle \nabla u_j, \nabla u_k \rangle}{\sum_{j,k=1}^{n} c_j c_k \langle u_j, u_k \rangle} = \frac{\sum_{j=1}^{n} \lambda_j c_j^2}{\sum_{j=1}^{n} c_j^2} \leq \frac{\sum_{j=1}^{n} \lambda_n c_j^2}{\sum_{j=1}^{n} c_j^2} = \lambda_n.$$

We've just shown that $\min_{v\in\Upsilon} \frac{||\nabla v||^2}{||v||^2} \leq \lambda_n$ for any arbitrary sequence $\{h_i\}_{i=1}^{n=1}$, and therefore

$$\max_{\{h_i\}_{i=1}^{n-1}} \min_{v\in\Upsilon} \frac{||\nabla v||^2}{||v||^2} \leq \lambda_n \tag{3.14}$$

also holds. To prove equality, we only need to consider the case where our arbitrary functions are the $n-1$ first eigenfunctions: $h_1 = u_1, ..., h_{n-1} = u_{n-1}$. By Theorem 3.2, we have for this choice of functions

$$\min_{v\in V^*} \frac{||\nabla v||^2}{||v||^2} = \lambda_n. \tag{3.15}$$

Since we now know that (3.15) holds for some choice of $h_1, ..., h_{n-1}$, then (3.14) must be an equality. □

The Maximin principle also holds for the Neumann eigenvalue problem, despite the lack of a boundary condition on the trial functions. The proof is nonetheless similar to the Dirichlet case and will therefore be omitted.

Theorem 3.6. *Consider the Neumann eigenvalue problem of the domain Ω and let $\tilde{h}_1, ..., \tilde{h}_{n-1}$ be arbitrary, piecewise continuous functions in Ω. Then the n-th eigenvalue $\tilde{\lambda}_n$ is given by*

$$\tilde{\lambda}_n = \max_{\{\tilde{h}_i\}_{i=1}^{n-1}} \min_{w\in\Psi} \frac{||\nabla w||^2}{||w||^2}, \tag{3.16}$$

where $\Psi := \left\{ w \in C^\infty(\overline{\Omega})\backslash\{0\} \mid \left\langle w, \tilde{h}_1 \right\rangle = ... = \left\langle w, \tilde{h}_{n-1} \right\rangle = 0 \right\}$ *defines the set of trial functions.*

3.3. Consequences of the Maximin principles

The Maximin principles are strong results that allow for eigenvalues of the Dirichlet eigenvalue problem to be directly compared to those of the Neumann eigenvalue problem. Moreover, the eigenvalues of *subdomains* $\Omega' \subset \Omega$ become comparable with the eigenvalues of the whole domain Ω. In what ways these comparisons work will become evident in this section.

3.3.1. Dirichlet vs Neumann

Theorem 3.7. *For the same domain Ω, consider any Dirichlet eigenvalue λ_i and any Neumann eigenvalue $\tilde{\lambda}_i$. Then*

$$\lambda_n \geq \tilde{\lambda}_n$$

for any fixed $n \in \mathbb{N}$.

In order to prove this theorem, we'll need the following lemma. This is a fundamental result that can be found in [11].

Lemma 3.2. *Let* A *and* B *be arbitrary, real sets such that* $B \subseteq A$. *Then*

$$\min A \leq \min B.$$

The idea is that if we consider two sets A and B to be defined in the same real space, but the elements in e.g. B are more restricted (say, by boundary conditions) compared to the elements in A such that $B \subset A$, then the smallest element of B is certainly no smaller than the smallest element of A.

Proof of theorem. Consider first the principal eigenvalues. Since $\tilde{\lambda}_1 = 0$ and $\lambda_1 > 0$, we have that $\lambda_1 > \tilde{\lambda}_1$.

For the n-th eigenvalues λ_n and $\tilde{\lambda}_n$, recall from the Maximin principles 3.5 and 3.6 that the sets Υ and Ψ of the Dirichlet and Neumann trial functions are given by

$$\Upsilon := \{v \in C_c^\infty(\Omega)\backslash\{0\} \mid \langle v, h_1\rangle = ... = \langle v, h_{n-1}\rangle = 0\},$$
$$\Psi := \left\{w \in C^\infty(\overline{\Omega})\backslash\{0\} \mid \left\langle w, \tilde{h}_1\right\rangle = ... = \left\langle w, \tilde{h}_{n-1}\right\rangle = 0\right\}.$$

Assume that $h_i = \tilde{h}_i \ \forall i = 1, ..., n-1$. Since the Neumann trial functions w don't have any boundary conditions, unlike the Dirichlet trial functions v, we can conclude that

$$\Upsilon \subset \Psi$$

By Lemma 3.2 and the Maximin principles, this yields

$$\lambda_n = \max_{\{h_i\}_{i=1}^{n-1}} \min_{v \in \Upsilon} \frac{||\nabla v||^2}{||v||^2} \geq \max_{\{h_i\}_{i=1}^{n-1}} \min_{w \in \Psi} \frac{||\nabla w||^2}{||w||^2} = \tilde{\lambda}_n.$$

□

3.3.2. Subdomains

Theorem 3.8. *Let* $\Omega' \subseteq \Omega$ *be any subdomain. Consider the Dirichlet eigenvalue problem for both domains with eigenvalues* λ_i *and* λ_i' *respectively. Then*

$$\lambda_n' \geq \lambda_n$$

for any fixed $n \in \mathbb{N}$.

Proof. Let $v' \in C_c^\infty(\Omega')\backslash\{0\}$ denote any trial function for Ω' and let $v \in C_c^\infty(\Omega)\backslash\{0\}$ denote any trial function for Ω. Assume both v', v are orthogonal to every h_i in a set of arbitrary piecewise continuous functions $\{h_i\}_{i=1}^{n-1}$ in Ω. Then the function v' can be defined in Ω in the following way:

$$v(x,y) = \begin{cases} v'(x,y) & (x,y) \in \Omega' \\ 0 & (x,y) \in \Omega\backslash\Omega'. \end{cases} \tag{3.17}$$

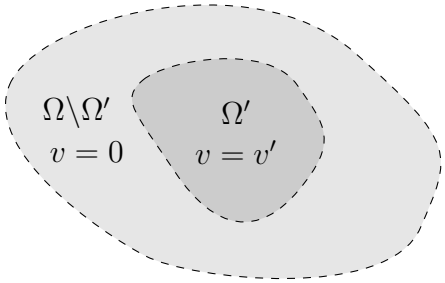


Figure 3.1: A visual example of (3.17).

Evidently, this v is still a trial function in Ω. This means that any trial function v' in Ω' is a trial function in Ω, but not conversely. The trial functions of Ω' thus have an additional constraint, which by similar argumentation to the previous proof means that

$$\lambda_n' = \max_{\{h_i\}_{i=1}^{n-1}} \min_{v' \in \Upsilon_{\Omega'}} \frac{||\nabla v'||^2}{||v'||^2} \geq \max_{\{h_i\}_{i=1}^{n-1}} \min_{v \in \Upsilon_{\Omega}} \frac{||\nabla v||^2}{||v||^2} = \lambda_n. \tag{3.18}$$

□

We will finalize this chapter with a theorem comparing *unions* of subdomains to the entire domain.

Theorem 3.9. *Let Ω be subdivided into a finite number of disjoint subdomains $\Omega_1, ..., \Omega_k \subset \Omega$ such that $\overline{\Omega}_1 \cup ... \cup \overline{\Omega}_k = \overline{\Omega}$ and each $\partial\Omega_1, ..., \partial\Omega_k$ is smooth. Consider the Laplacian eigenvalue problem for Ω and let $\{\lambda_i\}, \{\tilde{\lambda}_i\}$ be the ordered sequences of Dirichlet and Neumann eigenvalues respectively. Similarly, consider the Laplacian eigenvalue problem for each subdomain $\Omega_1, ..., \Omega_k$ and let $\{\mu_j\}, \{\tilde{\mu}_j\}$ be the combined ordered sequences of Dirichlet and Neumann eigenvalues respectively. Then*

$$\tilde{\mu}_n \leq \tilde{\lambda}_n \leq \lambda_n \leq \mu_n \tag{3.19}$$

for any fixed $n \in \mathbb{N}$.

Proof. Theorem 3.8 asserts that any eigenvalue $\mu_n \in \{\mu_j\}$ will be at least equal to the corresponding eigenvalue λ_n for all $n \in \mathbb{N}$. We thus have that

$$\lambda_n \leq \mu_n.$$

For the Neumann eigenvalues, recall from Theorem 3.6 that the trial functions w must be smooth in the interior. Since these trial functions do not have any boundary conditions, the ones related to the sequence $\{\tilde{\mu}_j\}$ are allowed to be *discontinuous* at the internal boundaries, whereas the trial functions related to $\{\tilde{\lambda}_i\}$ are necessarily continuous in all of $\overline{\Omega}$. With reasoning similar to the Dirichlet case, the extra restriction on the trial functions for the eigenvalues $\{\tilde{\lambda}_i\}$ yields the inequality

$$\tilde{\mu}_n \leq \tilde{\lambda}_n.$$

Note that $\tilde{\mu}_1 = \tilde{\lambda}_1 = 0$. Finally, by Theorem 3.7, we've proven the chain of inequalities given by (5.2.4) for any fixed $n \in \mathbb{N}$. □

This theorem will be of great importance in Chapter 5 for when we prove Weyl's law for any arbitrary domain Ω. However, we will also need the results of the Laplacian eigenvalue problem for some particular shapes known to have explicit solutions, so we'll explore those first in the next chapter.

We'll start by solving the Dirichlet eigenvalue problem for the rectangle.

4.1. The rectangle

Consider the domain given by $\Omega_R = \{(x, y) \in \mathbb{R}^2 \mid 0 < x < a,\ 0 < y < b,\ b \leq a\}$. This domain is precisely the interior of a rectangle in the xy-plane with side-lengths a and b (see Figure 4.1). Recall the problem (1.4) given as

$$\begin{cases} \Delta u + \lambda u = 0 & (x, y) \in \Omega_R \\ u = 0 & (x, y) \in \partial\Omega_R. \end{cases} \tag{4.1}$$

Substituting $u(x, y) = X(x)Y(y)$ into (4.1) and applying the boundary conditions in accordance to Ω_R gives

$$\begin{cases} \dfrac{d^2X(x)}{dx^2}Y(y) + X(x)\dfrac{d^2Y(y)}{dy^2} = -\lambda X(x)Y(y) & (x, y) \in ((0, a), (0, b)) \\ X(0)Y(y) = X(a)Y(y) = 0 & y \in [0, b] \\ X(x)Y(0) = X(x)Y(b) = 0 & x \in [0, a] \end{cases} \tag{4.2}$$

By dividing both sides of (4.2) with $X(x)Y(y)$, we get

$$\begin{cases} \dfrac{d^2X(x)}{dx^2}\dfrac{1}{X(x)} + \dfrac{d^2Y(y)}{dy^2}\dfrac{1}{Y(y)} = -\lambda & (x, y) \in ((0, a), (0, b)) \\ X(0) = X(a) = 0 \\ Y(0) = Y(b) = 0. \end{cases} \tag{4.3}$$

Since the terms on the LHS of the first equation of (4.3) depend on different variables, we can with the substitution $\lambda = \phi + \psi$ yield the following two systems of ODEs:

$$\begin{cases} \dfrac{d^2X(x)}{dx^2} + \phi X(x) = 0 & x \in [0, a] \\ X(0) = X(a) = 0 \end{cases} \qquad \begin{cases} \dfrac{d^2Y(y)}{dy^2} + \psi Y(y) = 0 & y \in [0, b] \\ Y(0) = Y(b) = 0. \end{cases}$$

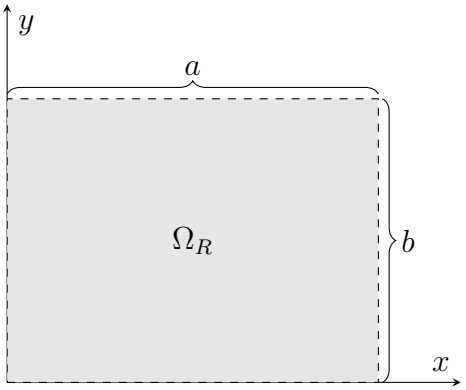


Figure 4.1

It can be easily verified that $\phi, \psi \leq 0$ only yields trivial solutions to these two systems, meaning that ϕ and ψ should be strictly positive (I'll leave it to the reader to carry out these calculations). Considering the left ODE, we can thus express $X(x)$ as

$$X(x) = c_1 cos(\sqrt{\phi}x) + c_2 sin(\sqrt{\phi}x),$$

where c_1, c_2 are constants. The boundary conditions give

$$\begin{aligned} X(0) = c_1 = c_1 cos(\sqrt{\phi}a) + c_2 sin(\sqrt{\phi}a) = X(a) = 0 \\ \implies c_1 = 0 \ \wedge \ c_2 sin(\sqrt{\phi}a) = 0 \\ \implies \sqrt{\phi} = l\frac{\pi}{a}, \quad l \in \mathbb{N}. \end{aligned}$$

This means that $\phi_l = l^2 \frac{\pi^2}{a^2}$ and (with $c_2 = 1$) $X_l(x) = sin(\frac{l\pi x}{a})$. Similarly, $\psi_m = m^2 \frac{\pi^2}{b^2}$ and $Y_m(y) = sin(\frac{m\pi y}{b})$ with $m \in \mathbb{N}$. Note that $l, m \neq 0$ since $\phi, \psi \neq 0$. We express the final results in the following proposition.

Proposition 4.1. *The eigenvalues and eigenfunctions for the Dirichlet eigenvalue problem of the domain* $\Omega_R = \{(x, y) \in \mathbb{R}^2 \mid 0 < x < a, \ 0 < y < b, \ b \leq a\}$ *are given explicitly by:*

$$\begin{cases} \lambda_{l,m} = l^2 \dfrac{\pi^2}{a^2} + m^2 \dfrac{\pi^2}{b^2} \\ u_{l,m} = sin\left(\dfrac{l\pi x}{a}\right) sin\left(\dfrac{m\pi y}{b}\right), \end{cases} \tag{4.4}$$

where $l, m \in \mathbb{N}$.

The Neumann eigenvalue problem for the rectangle also has explicit results. Since the process is much similar to the Dirichlet case, we'll only state the proposition and omit the calculations.

Proposition 4.2. *The eigenvalues and eigenfunctions for the Neumann eigenvalue problem of the domain* $\Omega_R = \{(x, y) \in \mathbb{R}^2 \mid 0 < x < a,\ 0 < y < b,\ b \leq a\}$ *are given explicitly by:*

$$\begin{cases} \tilde{\lambda}_{l,m} = l^2 \dfrac{\pi^2}{a^2} + m^2 \dfrac{\pi^2}{b^2} \\ \tilde{u}_{l,m} = cos\left(\dfrac{l\pi x}{a}\right) cos\left(\dfrac{m\pi y}{b}\right), \end{cases} \tag{4.5}$$

where $l, m \in \mathbb{N} \cup \{0\}$.

4.1.1. Comparing the rectangle with the square

If we consider the specific case $b = a$, we have the explicit solutions to the Dirichlet and Neumann problems for the square domain Ω_S with side a. Looking at the expression for $\lambda_{l,m}$ in (4.4) (and $\tilde{\lambda}_{l,m}$ in (4.5)), we can note that since $b \leq a$, we have

$$l^2 \frac{\pi^2}{b^2} \geq l^2 \frac{\pi^2}{a^2}, \quad \forall l \in \mathbb{N}.$$

This observation gives us the following Lemma:

Lemma 4.1. *Consider the Laplacian eigenvalue problem with either boundary condition for the square* Ω_S *with side* a *and the rectangle* Ω_R *with sides* a, b *such that* $b \leq a$ *respectively. Then the* n*-th eigenvalue for the square and the* n*-th eigenvalue for the rectangle satisfy*

$$\lambda_n(\Omega_S) \leq \lambda_n(\Omega_R)$$

for either boundary condition.

4.2. The isosceles right triangle

Consider the domain given by $\Omega_T = \{(x, y) \in \mathbb{R}^2 \mid 0 < y < x < a\}$, which describes an isosceles right triangle with side a in the xy-plane. Note that if the reflection of Ω_T with respect to the line $x = y$ is added together with the triangles hypotenuse, we get the square domain $\Omega_S = \{(x, y) \in \mathbb{R}^2 \mid 0 < x, y < a\}$. This symmetrical property allows us to make use of the explicit results for the rectangle in the previous section to find solutions for the isosceles right triangle as well.

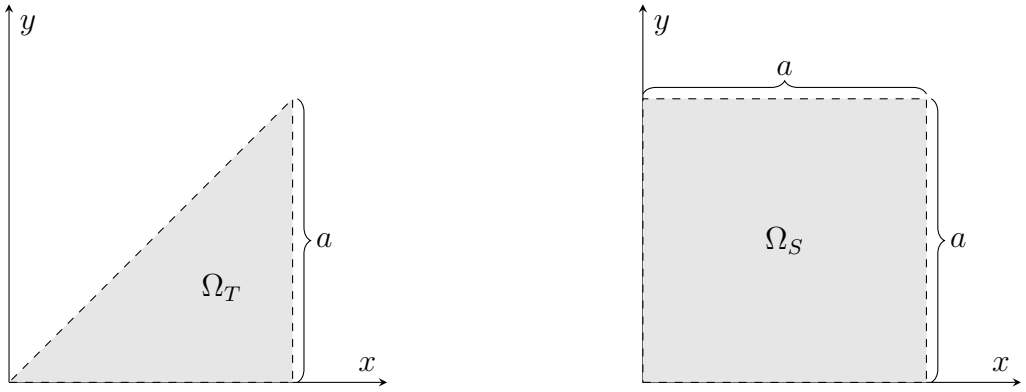


Figure 4.2

Consider the Dirichlet eigenvalue problem of the domain Ω_S, which by Proposition 4.1 has the following eigenfunctions/-values:

$$\begin{cases} \lambda_S = (l^2 + m^2)\dfrac{\pi^2}{a^2} \\ u_S = sin\left(\dfrac{l\pi x}{a}\right) sin\left(\dfrac{m\pi y}{a}\right), \end{cases} \tag{4.6}$$

where $l, m \in \mathbb{N}$. If $u_S(x, x) = 0 \ \forall x \in [0, a]$, then $u_S(x, y)$ restricted to Ω_T is also an eigenfunction of the Dirichlet eigenvalue problem for the triangle Ω_T denoted u_T with eigenvalue $\lambda_T = \lambda_S$. Conversely, given by the symmetric observation

$$(x, y) \in \overline{\Omega}_T \iff (y, x) \in \overline{\Omega_S \backslash \Omega_T},$$

we have that

$$u_S(x, y) = \begin{cases} u_T(x, y), & (x, y) \in \Omega_T \\ -u_T(y, x), & (x, y) \in \Omega_S \backslash \Omega_T \end{cases}$$

with eigenvalue $\lambda_S = \lambda_T$. Note that u_S necessarily changes sign on the different sides of the boundary $x = y$ since it has to remain smooth. We conclude that u_T and λ_T are found by taking combinations of u_S for the same eigenvalue λ_S such that $u_T = 0$ when $x = y$. This yields

$$\begin{aligned} u_T(x, y) &= u_S(x, y) - u_S(y, x) \\ &= sin\left(\frac{l\pi x}{a}\right) sin\left(\frac{m\pi y}{a}\right) - sin\left(\frac{l\pi y}{a}\right) sin\left(\frac{m\pi x}{a}\right), \end{aligned}$$

where $l, m \in \mathbb{N}$, $l < m$ and $(x, y) \in \Omega_T$. The restriction $l < m$ is imposed since $l = m$ would cause $u_T(x, y) = 0$ and since all pairs $(l, m), (m, l)$ would yield the same eigenfunction. We thus arrive at the following proposition.

Proposition 4.3. *The eigenvalues and eigenfunctions for the Dirichlet eigenvalue problem of the domain* $\Omega = \{(x, y) \in \mathbb{R}^2 \mid 0 < y < x < a\}$ *are given explicitly by:*

$$\begin{cases} \lambda_{l,m} = (l^2 + m^2)\dfrac{\pi^2}{a^2} \\ u_{l,m} = sin\left(\dfrac{l\pi x}{a}\right) sin\left(\dfrac{m\pi y}{a}\right) - sin\left(\dfrac{l\pi y}{a}\right) sin\left(\dfrac{m\pi x}{a}\right), \end{cases} \tag{4.7}$$

where $l, m \in \mathbb{N}$ *and* $l < m$.

With a similar argument, the eigenfunctions $\tilde{u}_T$ and eigenvalues $\tilde{\lambda}_T$ for the Neumann boundary condition of the isosceles right triangle are given by combinations of the eigenfunctions $\tilde{u}_S$ for the same eigenvalue $\tilde{\lambda}_S = \tilde{\lambda}_T$ of the square domain, but with the added boundary condition $\frac{\partial \tilde{u}_S}{\partial \vec{n}}(x, x) = 0 \ \forall x \in [0, a]$. From Proposition 4.2 we have that

$$\begin{cases} \tilde{\lambda}_S = (l^2 + m^2)\dfrac{\pi^2}{a^2} \\ \tilde{u}_S = cos\left(\dfrac{l\pi x}{a}\right) cos\left(\dfrac{m\pi y}{a}\right), \end{cases}$$

where $l, m \in \mathbb{N} \cup \{0\}$. And so, the eigenfunctions for Ω_T are given by

$$\begin{aligned} \tilde{u}_T(x, y) &= \tilde{u}_S(x, y) + \tilde{u}_S(y, x) \\ &= cos\left(\frac{l\pi x}{a}\right) cos\left(\frac{m\pi y}{a}\right) + cos\left(\frac{l\pi y}{a}\right) cos\left(\frac{m\pi x}{a}\right), \end{aligned}$$

where $l, m \in \mathbb{N} \cup \{0\}$ and $l \leq m$. Note that the unit outer normal $\vec{n}$ on the boundary $x = y$ is perpendicular to the hypotenuse of Ω_T, meaning that $\vec{n} = \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$. This leads to

$$\begin{aligned} \frac{\partial \tilde{u}_T}{\partial \vec{n}}(x, x) &= \nabla \tilde{u}_S(x, x) \cdot \vec{n} + \nabla \tilde{u}_S(x, x) \cdot \vec{n} \\ &= 2\left[\frac{\partial}{\partial x}\tilde{u}_S, \frac{\partial}{\partial x}\tilde{u}_S\right] \cdot \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right] = 0. \end{aligned}$$

This confirms that the added boundary condition $\frac{\partial \tilde{u}_S}{\partial \vec{n}}(x, x) = 0 \ \forall x \in [0, a]$ is satisfied, and because the other conditions are already met by the $\tilde{u}_S$ terms, we get the following proposition.

Proposition 4.4. *The eigenvalues and eigenfunctions for the Neumann eigenvalue problem of the domain* $\Omega = \{(x, y) \in \mathbb{R}^2 \mid 0 < y < x < a\}$ *are given explicitly by:*

$$\begin{cases} \tilde{\lambda}_{l,m} = (l^2 + m^2)\dfrac{\pi^2}{a^2} \\ \tilde{u}_{l,m} = cos\left(\dfrac{l\pi x}{a}\right) cos\left(\dfrac{m\pi y}{a}\right) + cos\left(\dfrac{l\pi y}{a}\right) cos\left(\dfrac{m\pi x}{a}\right), \end{cases} \tag{4.8}$$

where $l, m \in \mathbb{N} \cup \{0\}$ *and* $l \leq m$.

4.2.1. Comparing the isosceles right triangle with the square

The eigenvalues for either boundary condition of the isosceles right triangle Ω_T are more restricted than the ones related to the square Ω_S with side a. Similarly to the conclusions of Section 4.1.1, we get the following lemma.

Lemma 4.2. *Consider the Laplacian eigenvalue problem with either boundary condition for the square* Ω_S *with side* a *and the isosceles right triangle* Ω_T *with side* a *respectively. Then the* n*-th eigenvalue for the square and the* n*-th eigenvalue for the triangle satisfy*

$$\lambda_n(\Omega_S) \leq \lambda_n(\Omega_T)$$

for either boundary condition.

Now we are ready to tackle Weyl's law. We will prove it for both the rectangle Ω_R and any domain Ω based on methods found in [2].

5.1. The rectangle

Recall from Proposition 4.1 that the eigenfunctions and eigenvalues of the Dirichlet Laplacian for a rectangle are explicitly given by

$$\begin{cases} \lambda_{l,m} = l^2\dfrac{\pi^2}{a^2} + m^2\dfrac{\pi^2}{b^2} \\ u_{l,m} = sin\left(\dfrac{l\pi x}{a}\right) sin\left(\dfrac{m\pi y}{b}\right), \end{cases} \tag{5.1}$$

where $l, m \in \mathbb{N}$. Note that the equation for $\lambda_{l,m}$ describes the *quarter of an ellipse centered at the origin* in the first quadrant of the lm-plane (since $l, m > 0$). Note also that the eigenvalues $\lambda_{l,m}$ depend solely on the variables l and m. Since these variables are natural numbers, each eigenvalue corresponds to a positive lattice point in the lm-plane. Furthermore, since the eigenvalues form the non-decreasing sequence

$$0 < \lambda_1 < \lambda_2 \leq \ ... \ \leq \lambda_n \leq \ ...,$$

any lattice point $(l, m)_i$ uniquely associated to an eigenvalue $\lambda_i,\ i \in \{1, 2, ..., n\}$, lies *within* the closure of this quarter ellipse (see Figure 5.1) given an upper bound λ_n. Any such point is thus satisfying

$$\lambda_n \geq l^2\frac{\pi^2}{a^2} + m^2\frac{\pi^2}{b^2}.$$

With this conclusion, we additionally define $D(\lambda) :=$ *the number of eigenvalues not exceeding* λ *for the Dirichlet eigenvalue problem.* This means that $D(\lambda_n) = n$.

By observing Figure 5.1, it is clear that each marked lattice point can be represented by the upper right corner of a square whose interior is entirely contained within the

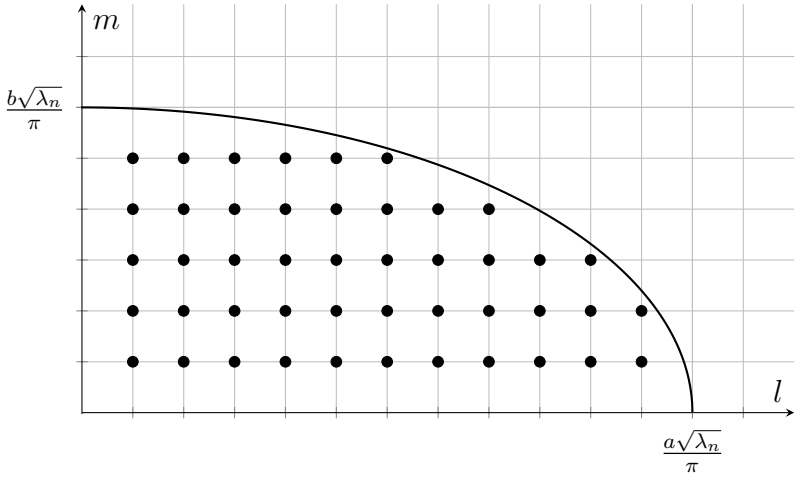


Figure 5.1: Each marked lattice point $(l, m)_i$ is associated to an eigenvalue λ_i. Note that the semi-axes depend on the value λ_n that is used as an upper bound.

quarter ellipse. The number $D(\lambda)$ will therefore never exceed its total area $\frac{\lambda ab}{4\pi}$. We can also conclude that $D(\lambda)$ and the area $\frac{\lambda ab}{4\pi}$ will differ approximately by the length of the perimeter of the quarter ellipse (which we'll denote $S(\lambda)$) for large λ. These observations yield the following chain of inequalities:

$$\frac{\lambda ab}{4\pi} - S(\lambda) \leq D(\lambda) \leq \frac{\lambda ab}{4\pi}. \tag{5.2}$$

Note that $S(\lambda)$ is of order $\sqrt{\lambda}$. In comparison to the term $\frac{\lambda ab}{4\pi}$, which is of order λ, $S(\lambda)$ will grow a lot slower and will for sufficiently large λ have a negligible effect on (5.2). This observation yields

$$D(\lambda) = \frac{ab}{4\pi}\lambda + \theta c\sqrt{\lambda}, \tag{5.3}$$

where c is a constant independent of λ and $|\theta| < 1$. This way of expressing $D(\lambda)$ will be used extensively in Section 5.2. By additionally making the substitution $\lambda = \lambda_n$, where λ_n is the n-th Dirichlet eigenvalue, we get

$$\lim_{n\to\infty} \frac{\lambda_n}{n} = \frac{4\pi}{ab}.$$

This describes an asymptotic relation between the spectra of a rectangle and its area ab and is precisely Weyl's law for the rectangle! The same result also holds for when the Neumann eigenvalue problem is considered with the only difference being that l and m are allowed to be 0. The proof of Weyl's law for the rectangle in [2] actually starts by proving it for the Neumann case and then makes the analogous connection to the Dirichlet case.

5.2. Any arbitrary domain Ω

We now know the asymptotic relation we wish to prove for any domain Ω. Let's express it here as a theorem:

Theorem 5.1. *Consider the Laplacian eigenvalue problem for any domain Ω with either boundary condition. Given an upper bound λ_n of the spectra $\sigma(\Omega)$ and the area $A(\Omega)$, the following asymptotic relation holds:*

$$\lim_{n\to\infty} \frac{\lambda_n}{n} = \frac{4\pi}{A(\Omega)}. \tag{5.4}$$

Weyl's law also proves that the sequences of eigenvalues for both the Dirichlet and the Neumann problem tend toward infinity. This relates back to what was established in Theorem 2.2.

We will now approximate any domain Ω by a collection of squares and right triangles and use the explicit results in the previous chapter to prove Weyl's law 5.1.

5.2.1. Finding upper bounds for some shapes

Let Ω_S denote a square and let Ω_T denote an isosceles right triangle, both of which have the same side-length a. By Lemma 4.2, we know that

$$\tilde{\lambda}_n(\Omega_S) \leq \tilde{\lambda}_n(\Omega_T) \tag{5.5}$$

for the n-th Neumann eigenvalues for the respective domains. Next, consider any arbitrary *right triangle* $\Omega_{T'}$ with sides a and b such that $b \leq a$. The goal is to make an appropriate transformation of $\Omega_{T'}$ into Ω_T and analyze how the Neumann eigenvalues are affected. In particular, we will see the transformation's effect on the Rayleigh quotient in (3.16) as well as the orthogonality property in Ψ from Theorem 3.6 when we apply it to the trial functions w. For $\Omega_{T'}$ with side a on the x-axis and side b on the y-axis, this Rayleigh quotient with $w = w(x, y)$ is given by

$$\frac{||\nabla w||^2}{||w||^2} = \frac{\langle \nabla w, \nabla w \rangle}{\langle w, w \rangle} = \frac{\iint_{\Omega_{T'}} \left[\left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2 \right] dxdy}{\iint_{\Omega_{T'}} w^2 \, dxdy}, \tag{5.6}$$

while the orthogonality property for $\tilde{h}_1, ..., \tilde{h}_{n-1}$ arbitrary, piecewise continuous functions in $\Omega_{T'}$ is given by

$$\left\langle w, \tilde{h}_i \right\rangle = \iint_{\Omega_{T'}} w\tilde{h}_i \, dxdy = 0 \quad \forall i \in \{1, ..., n-1\}. \tag{5.7}$$

We then make the following substitution to transform $\Omega_{T'}$ into Ω_T in the $\xi\eta$-plane.

$$\begin{cases} \xi = x \\ \eta = a\dfrac{y}{b} \end{cases} \implies dxdy = \left|\frac{\partial(\xi,\eta)}{\partial(x,y)}\right|^{-1} d\xi d\eta = \frac{b}{a} d\xi d\eta. \tag{5.8}$$

We can confirm straight away that this transformation does not affect the orthogonality property of (5.7) since $\frac{b}{a} \neq 0$.

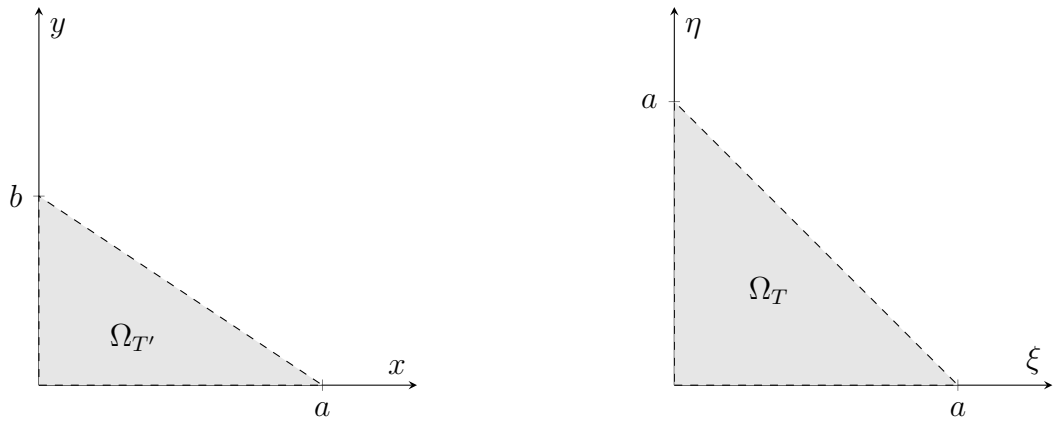


Figure 5.2

By inserting (5.8) into (5.6), we get

$$\begin{aligned} \frac{\iint_{\Omega_{T'}} \left[\left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2\right] dxdy}{\iint_{\Omega_{T'}} w^2\, dxdy} &= \frac{\iint_{\Omega_T} \left[\left(\frac{\partial w}{\partial \xi}\right)^2 + \frac{a^2}{b^2}\left(\frac{\partial w}{\partial \eta}\right)^2\right] \frac{b}{a}\, d\xi d\eta}{\iint_{\Omega_T} w^2 \frac{b}{a}\, d\xi d\eta} \\ &= \frac{\iint_{\Omega_T} \left[\left(\frac{\partial w}{\partial \xi}\right)^2 + \frac{a^2}{b^2}\left(\frac{\partial w}{\partial \eta}\right)^2\right] d\xi d\eta}{\iint_{\Omega_T} w^2\, d\xi d\eta}. \end{aligned} \tag{5.9}$$

Since $b \leq a \implies \frac{a}{b} \geq 1$, we get the following inequality:

$$\frac{\iint_{\Omega_T} \left[\left(\frac{\partial w}{\partial \xi}\right)^2 + \frac{a^2}{b^2}\left(\frac{\partial w}{\partial \eta}\right)^2\right] d\xi d\eta}{\iint_{\Omega_T} w^2\, d\xi d\eta} \geq \frac{\iint_{\Omega_T} \left[\left(\frac{\partial w}{\partial \xi}\right)^2 + \left(\frac{\partial w}{\partial \eta}\right)^2\right] d\xi d\eta}{\iint_{\Omega_T} w^2\, d\xi d\eta}. \tag{5.10}$$

Note that the RHS of this inequality is the Rayleigh quotient in the same sense as (5.6), but for the isosceles right triangle Ω_T. Thus, by taking the maximum of the

minimum on both sides of (5.10) in accordance to the Neumann Maximin principle 3.6, we can conclude together with (5.5) that

$$\tilde{\lambda}_n(\Omega_S) \leq \tilde{\lambda}_n(\Omega_T) \leq \tilde{\lambda}_n(\Omega_{T'}). \tag{5.11}$$

And similarly for the rectangle Ω_R with sides a and b such that $b \leq a$, we have by Lemma 4.1:

$$\tilde{\lambda}_n(\Omega_S) \leq \tilde{\lambda}_n(\Omega_R) \tag{5.12}$$

Now, let the notation $N_i(\lambda)$ represent the number of eigenvalues not exceeding some fixed $\lambda > 0$ for the Neumann eigenvalues of any domain $\Omega_i \subseteq \Omega_S$. Together with the results of (5.11) and (5.12), we get

$$\left.\begin{aligned} N_{T'}(\lambda) \leq N_T(\lambda) \\ N_R(\lambda) \end{aligned}\right\} \leq N_S(\lambda), \tag{5.13}$$

where λ acts as an upper bound for all the different domain's spectra. These results, together with Theorem 3.9, conclude that it is possible for any domain Ω comprised of a finite number of rectangles and right triangles to find an upper bound for the number of eigenvalues less than some given bound.

5.2.2. Approximating Ω

We start by subdividing the domain Ω into squares of side a. The partition should be fine enough such that, given any part of the boundary $\partial\Omega$ contained in any of these squares, the direction of the normal of the boundary strip doesn't vary by more than some small angle α.

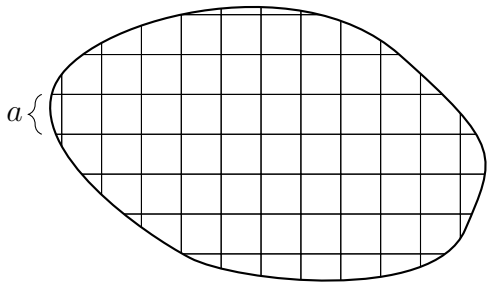


Figure 5.3: The domain Ω partitioned as $E_1 \cup ... \cup E_r \cup Q_1 \cup ... \cup Q_k$. Note that the length a should ideally be much smaller and is enlarged here for visual purposes.

Now, let $r \in \mathbb{N}\backslash\{1\}$ be the number of adjacent "squares" $E_1, ..., E_r$ that are intersected and consequently deformed by the boundary $\partial\Omega$. With respect to rotation

and reflection, there are only two different ways the domains E_i are structured for sufficiently small α (see Figure 5.4). If the boundary strip $E_1 \cup ... \cup E_r$ is removed from Ω, we end up with a domain completely exhausted by k squares $Q_1, ..., Q_k$. Note also that

$$r < \frac{C}{a}, \tag{5.14}$$

where C is some constant depending on the length of the boundary $\partial\Omega$ (which, in turn, is independent of a).

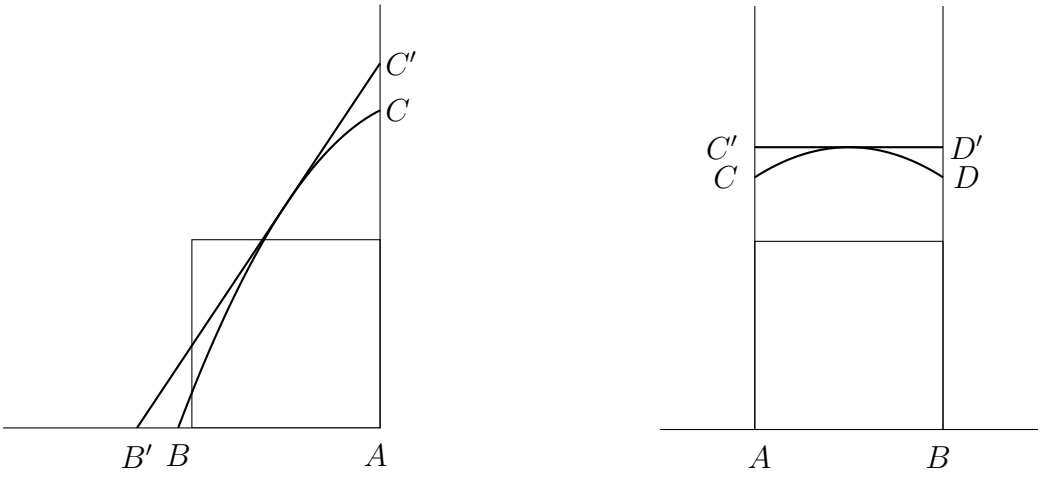


Figure 5.4

Let's now look at the different types of domains E_i. We will refer to Figure 5.4a and Figure 5.4b as E_i^1 and E_i^2 respectively.

E_i^1 is bounded by two perpendicular line-segments AB and AC with lengths between a and $3a$ and a segment BC of the boundary. E_i^2 is bounded by a line-segment AB of length a, two line-segments AC and BD perpendicular to AB with lengths between a and $3a$ and a segment CD of the boundary. The straight boundaries of any E_i are composed of edges of adjacent domains of the partition. This construction might not be too evident, but we will nonetheless not give it a rigorous justification here. Further comments regarding this can be found in [2].

Next, we'll choose any arbitrary point on BC in E_i^1 and on CD in E_i^2 respectively and draw tangents through them. These are represented by $B'C'$ and $C'D'$ in Figure 5.4. For sufficiently small α, we get a right triangle $AB'C'$ and a trapezoid $ABC'D'$, both

of which have sides smaller than $4a$ (this follows from the fact that the length of any straight boundary of any E_i doesn't exceed $3a$). We will denote any of these two variants as E_i'.

Now we want to make appropriate transformations, deforming any E_i into some E_i', and see how the n-th eigenvalue for the Neumann eigenvalue problem for either domain is affected. Let's start with E_i^1: We'll use polar coordinates ρ, θ, to express the different line-segments. If A denotes the pole, $\rho = f(\theta)$ denotes BC and $\rho = g(\theta)$ denotes $B'C'$, then the transformation given by

$$\begin{cases} \theta' = \theta \\ \rho' = \rho \dfrac{g(\theta)}{f(\theta)} \end{cases} \tag{5.15}$$

transforms the curvilinear triangle ABC into the rectilinear triangle $AB'C'$. For E_i^2 we will use regular cartesian coordinates. If AB lies along the x-axis, $y = f(x)$ denotes CD and $y = g(x)$ denotes $C'D'$, then the transformation given by

$$\begin{cases} x' = x \\ y' = y \dfrac{g(x)}{f(x)} \end{cases} \tag{5.16}$$

transforms the domain $ABCD$ into the trapezoid $ABC'D'$.

5.2.3. The effect of the transformations on the eigenvalues

The transformations of E_i into E_i' given by (5.15) and (5.16) do, in fact, have certain properties that imply that the n-th Neumann eigenvalue $\tilde{\lambda}_n$ differs arbitrarily little from $\tilde{\lambda}_n'$ for sufficiently small α. To see when and how this happens, we'll start by providing a definition based on results found in [2].

Definition 5.1. Given two domains $G, G' \subsetneq \mathbb{R}^2$ such that $\overline{G}$ is transformed pointwise into $\overline{G'}$ by equations of the form

$$\begin{cases} x' = x + f(x, y) \\ y' = y + g(x, y), \end{cases} \tag{5.17}$$

where f, g are continuous with piecewise continuous first derivatives in the domain, we say that the domain G is approximated by the domain G' with the degree of accuracy ϵ if

$$|f|, |g|, \left|\frac{\partial f}{\partial x_i}\right|, \left|\frac{\partial g}{\partial x_i}\right| < \epsilon,$$

where $x_i \in \{x, y\}$ and $\epsilon > 0$. When $\epsilon \to 0$, we say that G' *is deformed continuously into* G.

Comparing this definition with the transformations (5.15) and (5.16), it becomes evident that for sufficiently fine partitions of Ω (and thus sufficiently small values of α) these transformations have the same form as (5.17) with ϵ being sufficiently small. Let's pair this observation with the following theorem:

Theorem 5.2. *Consider the Neumann eigenvalue problem for the domains G and G' respectively. If G is deformed into G' when $\epsilon \to 0$ in the sense given by Definition 5.1, then*

$$\left| \frac{\tilde{\lambda}'_n}{\tilde{\lambda}_n} - 1 \right| < \beta, \tag{5.18}$$

where $\beta \to 0$ as $\epsilon \to 0$ and $\tilde{\lambda}'_n, \tilde{\lambda}_n$ are the respective domain's n-th Neumann eigenvalues.

Remark 5.1. This theorem is a simpler version of a theorem found in [2], where even the Dirichlet case is considered. Since we only need the result for the Neumann case, we'll omit the part of the proof that handles the boundary condition (since the Neumann trial functions don't have any boundary conditions).

Proof. Similarly to the transformation process in Section 5.2.1, we will work with the Rayleigh quotient in (3.16) and the orthogonality property in Ψ from Theorem 3.6. For the domain G and $w = w(x, y)$ this Rayleigh quotient is given by

$$\frac{\iint_{\Omega_G} \left[\left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2 \right] dxdy}{\iint_{\Omega_G} w^2 \, dxdy}, \tag{5.19}$$

while the orthogonality property is given by $\left\langle w, \tilde{h}_i \right\rangle = 0$ for all arbitrary, piecewise continuous functions $\tilde{h}_1, ..., \tilde{h}_{n-1}$ in G, similar to (5.7). Assume the transformations are given by (5.17). This yields

$$dxdy = \left| \frac{\partial(x', y')}{\partial(x, y)} \right|^{-1} dx'dy', \qquad \begin{cases} \dfrac{\partial w}{\partial x} = \dfrac{\partial w}{\partial x'} \left(1 + \dfrac{\partial f}{\partial x}\right) + \dfrac{\partial w}{\partial y'} \dfrac{\partial g}{\partial x} \\ \dfrac{\partial w}{\partial y} = \dfrac{\partial w}{\partial y'} \left(1 + \dfrac{\partial g}{\partial y}\right) + \dfrac{\partial w}{\partial x'} \dfrac{\partial f}{\partial y}. \end{cases}$$

Substituting this transformation into (5.19) gives

$$\frac{\iint_{\Omega_{G'}} \left[\left(\frac{\partial w}{\partial x'} \left(1 + \frac{\partial f}{\partial x} \right) + \frac{\partial w}{\partial y'} \frac{\partial g}{\partial x} \right)^2 + \left(\frac{\partial w}{\partial y'} \left(1 + \frac{\partial g}{\partial y} \right) + \frac{\partial w}{\partial x'} \frac{\partial f}{\partial y} \right)^2 \right] M^{-1} dx' dy'}{\iint_{\Omega_{G'}} w^2 M^{-1} dx' dy'}, \tag{5.20}$$

where $M = \left(1 + \frac{\partial f}{\partial x}\right)\left(1 + \frac{\partial g}{\partial y}\right) - \frac{\partial f}{\partial y}\frac{\partial g}{\partial x}$. For the same M, the orthogonality property becomes

$$\iint_{\Omega_G} w\tilde{h}_i \, dxdy = \iint_{\Omega_{G'}} w\tilde{h}_i M^{-1} dx' dy'. \tag{5.21}$$

If we let $\epsilon \to 0$, then $|f|, |g|, \left|\frac{\partial f}{\partial x_i}\right|, \left|\frac{\partial g}{\partial x_i}\right| \to 0$ by Definition 5.1. Additionally, $M \to 1$ by this observation. We can therefore assume that when $\epsilon \to 0$, (5.20) goes toward

$$\frac{\iint_{\Omega_{G'}} \left[\left(\frac{\partial w}{\partial x'} \right)^2 + \left(\frac{\partial w}{\partial y'} \right)^2 \right] dx' dy'}{\iint_{\Omega_{G'}} w^2 dx' dy'} \tag{5.22}$$

and the orthogonality property is preserved in (5.21). By the Maximin principle 3.6, we can conclude that $\tilde{\lambda}_n$ differs arbitrarily little from $\tilde{\lambda}'_n$ during this deformation. In particular, the inequality (5.18) holds for some small β such that $\beta \to 0$ as $\epsilon \to 0$. □

Consequently, the eigenvalues $\tilde{\lambda}_n$ and $\tilde{\lambda}'_n$ for the domains E_i and E'_i differ arbitrarily little for sufficiently small a. We can thus allow ourselves to utilize the spectra of E'_i to draw conclusions about the spectra of E_i.

Recall by (5.13) that

$$\left.\begin{aligned} N_{T'}(\lambda) \leq N_T(\lambda) \\ N_R(\lambda) \end{aligned}\right\} \leq N_S(\lambda) \tag{5.23}$$

for some upper bound $\lambda > 0$. Similarly to (5.3) from Section 5.1, we can express $N_S(\lambda)$ as

$$N_S(\lambda) = \frac{a^2\lambda}{4\pi} + a\theta_1 c_1 \sqrt{\lambda}, \tag{5.24}$$

where $|\theta_1| < 1$ and c_1 is independent of a and λ (note that a appears here in both terms. This is because all sides of the square are of length a, meaning that the perimeter length $S(\lambda)$ necessarily has a factor a). Since the domains E'_i consist of either a right triangle with sides smaller than $4a$ or a combination of such a triangle

and a rectangle with sides smaller than $3a$, we can by (5.23) find constants c_2, c_3 independent of a and λ such that

$$N_{E_i'}(\lambda) = c_2 a^2 \lambda + c_3 a\sqrt{\lambda},$$

where $N_{E_i'}(\lambda)$ is the number of Neumann eigenvalues of any domain E_i' not exceeding λ. By (5.18) for sufficiently small β (i.e. sufficiently small a), we can thus conclude that

$$N_{E_i}(\lambda) = \theta_2(c_2 a^2 \lambda + c_3 a\sqrt{\lambda}), \tag{5.25}$$

where $0 < \theta_2 < 1$.

5.2.4. Assembling the final chain of inequalities

Fix some $\lambda > 0$ and let $D(\lambda)$ represent the number of Dirichlet eigenvalues not exceeding λ of the entire domain Ω, and equivalently $D_i(\lambda)$ for any domain Q_i. Let $N_i(\lambda)$ represent the number of Neumann eigenvalues not exceeding λ of any domain Q_i, and equivalently $N_{E_i}(\lambda)$ for any domain E_i. Since we know by Theorem 3.9 that

$$\tilde{\mu}_n \leq \tilde{\lambda}_n \leq \lambda_n \leq \mu_n,$$

we can conclude that

$$\sum_{i=1}^{k} D_i(\lambda) \leq D(\lambda) \leq \sum_{i=1}^{k} N_i(\lambda) + \sum_{i=1}^{r} N_{E_i}(\lambda). \tag{5.26}$$

We'll start by finding the lower bound of (5.26). By (5.24), we get

$$\sum_{i=1}^{k} D_i(\lambda) = k\left(\frac{a^2\lambda}{4\pi} + a\theta_1 c_1\sqrt{\lambda}\right), \tag{5.27}$$

where $|\theta_1| < 1$ and c_1 is independent of a, k and λ. For the upper bound of (5.26), we can deduce from (5.24) and (5.25) that

$$\sum_{i=1}^{k} N_i(\lambda) + \sum_{i=1}^{r} N_{E_i}(\lambda) = k\left(\frac{a^2\lambda}{4\pi} + a\theta_2 c_2\sqrt{\lambda}\right) + r\theta_3\left(a^2 c_3\lambda + a c_4\sqrt{\lambda}\right), \tag{5.28}$$

where $|\theta_2| < 1$, $0 < \theta_3 < 1$ and every c_i is independent of a, k, r and λ.

We can now assemble our final chain of inequalities. By combining (5.26), (5.27) and (5.28), we get

$$k\left(\frac{a^2\lambda}{4\pi} + a\theta_1 c_1\sqrt{\lambda}\right) \leq D(\lambda) \leq k\left(\frac{a^2\lambda}{4\pi} + a\theta_2 c_2\sqrt{\lambda}\right) + r\theta_3\left(a^2 c_3\lambda + a c_4\sqrt{\lambda}\right) \tag{5.29}$$

By dividing (5.29) with λ and then applying the substitution $\lambda = \lambda_n$, where λ_n is the n-th Dirichlet eigenvalue for the entire domain Ω, we get

$$\left(\frac{ka^2}{4\pi}\right) + (ka\theta_1 c_1)\frac{1}{\sqrt{\lambda_n}} \leq \frac{n}{\lambda_n} \leq \left(\frac{ka^2}{4\pi} + ra^2\theta_3 c_3\right) + (ka\theta_2 c_2 + ra\theta_3 c_4)\frac{1}{\sqrt{\lambda_n}} \quad (5.30)$$

Now, recall from (5.14) that $r < \frac{C}{a}$, and so $ra < C$. By taking a sufficiently small *fixed* a, we therefore have that ra^2 is arbitrarily small. Note also that

$$|ka^2 - A(\Omega)| < \delta$$

for arbitrarily small δ. And so, by firstly taking the limit of (5.30) as $n \to \infty$, we obtain

$$\frac{ka^2}{4\pi} \leq \lim_{n\to\infty} \frac{n}{\lambda_n} \leq \left(\frac{ka^2}{4\pi} + ra^2\theta_3 c_3\right) \quad (5.31)$$

We here follow a similar argumentation as in Section 5.1, where the terms of order $\sqrt{\lambda}$ have a negligible effect for large λ. By secondly taking the limit of (5.31) as $a \to 0$, k will grow significantly faster than r, meaning that $ra^2 \to 0$ and $\delta \to 0$ in this context. Because λ_n and n are independent of a, we therefore get

$$\frac{A(\Omega)}{4\pi} \leq \lim_{n\to\infty} \frac{n}{\lambda_n} \leq \frac{A(\Omega)}{4\pi}$$
$$\Longleftrightarrow$$
$$\lim_{n\to\infty} \frac{\lambda_n}{n} = \frac{4\pi}{A(\Omega)},$$

proving Weyl's law for the Dirichlet eigenvalue problem of the domain Ω. However, since $\tilde{\lambda}_n$ is *also* squeezed in between μ_n and $\tilde{\mu}_n$ by Theorem 3.9, this additionally implies that Weyl's law holds for when the Neumann eigenvalue problem of Ω is considered. Therefore, this proves all of Theorem 5.1. Although the homogeneous Neumann boundary condition is not related to the physical nature of drums as far as we know, this still sees usage in other areas of physics/mathematics.

5.3. Comment on the absence of vertices

In Definition 2.1 we let the boundary $\partial\Omega$ be smooth in $\mathbb{R}^2$, but what if we allowed for it to have a finite number of vertices? It can be shown that Weyl's law, in fact, still holds for these types of domains [2]. I've nonetheless decided not to include such boundaries in this thesis as it would require additional approximations for re-entrant vertices in particular (note that the explicit results of Chapter 4 only handles figures with salient vertices). More details regarding this can be found in [3].

6. SUMMARY

When one is deeply invested in mathematical details of an originally physical problem, it is easy to forget what practical implementation a result might yield. This thesis views the problem through very theoretical lenses and has not discussed some relevant physical aspects that might affect the result. For example, since Weyl's law considers very large eigenvalues, it would require the striker of the drum to input a sufficiently large amount of energy to excite a proper amount of pure tones to yield an acceptable approximation of the area. Moreover, the hearing range of the human ear is most likely not vast enough to actually "hear the area of a drum" anyway. However, Weyl's law and its extension to compact Riemannian manifolds have played an important role in modeling other areas of science, especially in the context of black body radiation. For anyone intrigued to learn more, see [1], where the history of Weyl's law and its importance in physics are well documented.

A. APPENDIX

A.1. Gauss-Green Theorem

Let $\Omega \subsetneq R^n$ be open and bounded with a smooth boundary $\partial\Omega$. Suppose $u \in C^1(\Omega) \cap C(\overline{\Omega})$. Then

$$\int_\Omega \frac{\partial u}{\partial x_i}\, dx = \int_{\partial\Omega} u\vec{n}_i\, dS,$$

where $i = 1, ..., n$. This implies for each vector field $\mathbf{u} \in C^1(\Omega; R^n) \cap C(\overline{\Omega}; \mathbb{R}^n)$ that

$$\int_\Omega \nabla \cdot \mathbf{u}\, dx = \int_{\partial\Omega} \mathbf{u} \cdot \vec{n}\, dS.$$

Source: [3].

A.2. Gram-Schmidt process

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space and let $u_1, ..., u_n \in V$ be n linearly independent vectors. If $v_1, ..., v_k$ for $k < n$ forms an orthogonal basis of span$\{u_1, ..., u_k\}$, then $v_1, ..., v_{k+1}$ forms an orthogonal basis of span$\{u_1, ..., u_{k+1}\}$, where

$$v_{k+1} = u_{k+1} - \sum_{i=1}^{k} \frac{\langle u_{k+1}, v_i\rangle}{\langle v_i, v_i\rangle} v_i,$$

and $e_1, ..., e_{k+1}$ forms an orthonormal basis of span$\{u_1, ..., u_{k+1}\}$, where

$$e_j = \frac{v_j}{||v_j||}, \quad j = 1, ..., k+1.$$

Source: [7].

A.3. Compact support

Let $\Omega \subsetneq R^n$ be bounded. The *support* of a function $f(x_1, ..., x_n) \in C(\Omega)$ is defined as

$$\text{supp}(f(x_i)) := \{x_i \in \Omega \mid f(x_i) \neq 0\}.$$

If the support of f is a compact subset of Ω, then we say that f has *compact support* in Ω and write $f \in C_c(\Omega)$. This means that the function f varies continuously to the boundary $\partial\Omega$ such that $f|_{\mathbb{R}^n \setminus \Omega} = 0$.

Source: [9].

Notes on the usage of artificial intelligence

ChatGPT (URL: `https://chatgpt.com/`) has been used extensively as a guide to code the various figures of this thesis. It has also been used to translate some mathematical terms from Swedish to English.

REFERENCES

[1] W. Arendt and W.P. Schleich. *Mathematical Analysis of Evolution, Information and Complexity*. John Wiley & Sons, 2009.

[2] R. Courant and D. Hilbert. *Methods of Mathematical Physics Vol. 1*. John Wiley & Sons, 1953.

[3] L. C. Evans. *Partial Differential Equations*. American Mathematical Society, 2 edition, 2010.

[4] I. M. Gel'fand and S. V. Fomin. *Calculus of variations*. Selected Russian publications in the mathematical sciences. Prentice-Hall, rev. english edition, 1963.

[5] C. Gordon, D. L. Webb, and S. Wolpert. One cannot hear the shape of a drum. *Bulletin (new series) of the American Mathematical Society*, 27(1):134–138, 1992.

[6] M. Kac. Can one hear the shape of a drum? *The American Mathematical Monthly*, 73(4P2), 1966.

[7] Herschend M. and Kragh T. Linjär algebra II, 2023.

[8] Y. Pinchover and J. Rubinstein. *An Introduction to Partial Differential Equations*. Cambridge university press, 2005.

[9] Wolfram Research. Compact support. `https://mathworld.wolfram.com/CompactSupport.html`. Accessed: June 12, 2025.

[10] K. F. Riley. *Mathematical Methods for Physics and Engineering*. Cambridge University Press, 2006.

[11] W. Rudin. *Principles of Mathematical Analysis*. McGraw Hill, 3 edition, 1976.

[12] W. A. Strauss. *Partial Differential Equations: An Introduction*. John Wiley & Sons, 2 edition, 2007.

[13] H. Weyl. Über die asymptotische verteilung der eigenwerte. *Nachrichten von der Königlichen Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1911.

[14] N. Young. *An Introduction to Hilbert Space*. Cambridge mathematical textbooks. Cambridge University Press, 1988.